



Technical University of Applied Sciences
Würzburg-Schweinfurt

Basics of Electronics and Components

3. Magnetostatics and induction

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Chapter outline

Learning objectives Students will be able to: - Identify the four Maxwell equations in differential and integral form for systems in equilibrium - Explain the physical meaning of each equation (Gauss' law, conservative fields, no magnetic monopoles, already covered) - Apply Gauss's and Stokes' theorems to convert between integral and differential form - Distinguish between electrostatics and magnetostatics equations 3.1 Maxwell's equations Basics of electronics with components © Prof. Dr. Henning Lormann, 2026	Learning objectives Students will be able to: - Summarize important properties of magnetic fields - Calculate the magnetic force on a moving charge - Describe the difference between magnetic field intensity and flux density - Describe the concept of magnetic flux - Name Biot-Savart law of magnetostatics and derive why there are no magnetic monopolies - Explain the similarities and differences between electric and magnetic fields 3.2 Basic concepts of magnetic fields Basics of electronics with components © Prof. Dr. Henning Lormann, 2026	Learning objectives Students will be able to: - Explain the nature of magnetic fields around current carrying wires - Determine the shape of the magnetic field around different wire layouts - Explain the concept and formulation of the magnetic voltage and the magnetomotive force (MMF) - Apply Ampere's circuital law to calculate the magnetic field for basic setups 3.3 Stationary magnetic fields Basics of electronics with components © Prof. Dr. Henning Lormann, 2026	Learning objectives Students will be able to: - Explain the causes for magnetization of matter - Explain the magnetic flux density to the magnetic field intensity using magnetization curves - Extract the relative permeability from magnetization curves - Explain magnetic hysteresis - Select the right core material for your applications 3.4 Magnetization of matter Basics of electronics with components © Prof. Dr. Henning Lormann, 2026
Learning objective Students will be able to: - Explain the concept of magnetic circuits - Apply the concepts of magnetic flux and magnetomotive force to calculate magnetic circuits Note: Throughout this chapter we use the <i>passive</i> convention $i \leftarrow \int \vec{J} d\vec{A}$ and $U \leftarrow \int \vec{E} d\vec{s}$, i.e. i and U denote the magnitudes of the corresponding vector fields 3.5 Magnetic circuit Basics of electronics with components © Prof. Dr. Henning Lormann, 2026	Learning objective Students will be able to: - Explain Maxwell's second equation / Faraday's law of induction - Determine the voltage induced by a changing magnetic field through a loop of wire - Describe the self-induction of an electrical conductor - Calculate the energy stored in a magnetic field Note: When talking about currents and voltages in this chapter, they might be constant or time-varying, dependent of the task of change of the magnetic flux. We therefore utilize small letters i and u. 3.6 Time varying fields / Electromagnetic induction Basics of electronics with components © Prof. Dr. Henning Lormann, 2026	Learning objective Students will be able to: - Explain the working principle of an ideal transformer - Apply your knowledge on electromagnetic induction and Ampere's circuital law - Apply the dot convention to define the phase of the secondary circuit - Calculate the parameters (voltage, current, turn ratio, reflected impedance) of ideal transformer 3.7 Ideal transformer Basics of electronics with components © Prof. Dr. Henning Lormann, 2026	

3. Magnetostatics and induction

3. Magnetostatics and induction

Foreword

- The fundamental idea of magnetism is the concept of a **magnetic field exerting a force on moving electric charges**.
- Many of the **concepts of electrostatics can be transferred** surprisingly easily and show a beautiful symmetry. In other cases, the behaviour is just opposite to what you'd expect from electric fields.
- This chapter will mainly cover magnetostatics, **non-varying magnetic fields: Charges** may move at a **constant velocity** and **constant current** may flow.
- To introduce induction and electric machines, the chapter briefly addresses time-varying magnetic fields and electromagnetic induction.

Learning objectives

Students will be able to:

- Identify the four Maxwell equations in differential and integral forms for systems in equilibrium
- Explain the physical meaning of each equation (Gauss' law, conservative fields, no magnetic monopoles, steady currents)
- Apply Gauss's and Stokes' theorems to convert between integral and differential forms
- Distinguish between electrostatic and magnetostatic equations

3.1 Maxwell's equations

3.1 Maxwell's equations

Differential form

	Formula	Name and physical meaning
I.	$\vec{\nabla} \cdot \vec{D} = \rho$	Gauss' law: “The net electric flux $\vec{\nabla} \cdot \vec{D}$ is proportional to the charge density ρ at that same point.”
II.	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	Faraday's law of induction: “a changing magnetic field $\frac{\partial \vec{B}}{\partial t}$ creates a circulating electric field $\vec{\nabla} \times \vec{E}$.”
III.	$\vec{\nabla} \cdot \vec{B} = 0$	Gauss' law of magnetics: “the magnetic field has zero divergence, meaning magnetic field lines never begin or end at a point (no isolated magnetic poles).”
IV.	$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	Ampère–Maxwell equation: “current density $\vec{J}$ and a time-varying flux density $\frac{\partial \vec{D}}{\partial t}$ and produces a magnetic field $\vec{\nabla} \times \vec{H}$.”

3.1 Maxwell's equations

Systems in equilibrium

Formula

Name and physical meaning

Electrostatics

I. $\vec{\nabla} \cdot \vec{D} = \rho$

Gauss' law: "The net electric flux $\vec{\nabla} \cdot \vec{D}$ is proportional to the charge density ρ at that same point."

II. $\vec{\nabla} \times \vec{E} = 0$

Limit of Faraday's law: "In equilibrium there is no induced electric field." (conservative field)

Magnetostatics

III. $\vec{\nabla} \cdot \vec{B} = 0$

Gauss' law of magnetics: "There are no free magnetic poles."

IV. $\vec{\nabla} \times \vec{H} = \vec{j}$

Ampère's law: "Electric currents create steady magnetic field loops around them." (right hand rule)

3.1 Maxwell's equations

Comparing integral and differential form in equilibrium

Integral form

I. $\oiint_{\partial\Omega} \vec{D} \cdot d\vec{A} = \iiint_{\Omega} \vec{\nabla} \cdot \vec{D} dV = Q_{\text{encl.}}$

II. $\oint_{\partial\Sigma} \vec{E} \cdot d\vec{l} = \iint_{\Sigma} \vec{\nabla} \times \vec{E} \cdot d\vec{A} = 0$

III. $\oiint_{\partial\Omega} \vec{B} \cdot d\vec{A} = \iiint_{\Omega} \vec{\nabla} \cdot \vec{B} dV = 0$

IV. $\oint_{\partial\Sigma} \vec{H} \cdot d\vec{l} = \iint_{\Sigma} \vec{\nabla} \times \vec{H} \cdot d\vec{A} = I$

Differential form

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \times \vec{E} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

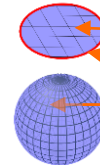
$$\vec{\nabla} \times \vec{H} = \vec{j}$$



Using Gauss's and Stokes's theorems, we can convert between integral and differential forms. See chapter 1 for the mathematical details.

Gauss' law

Stokes' law



Σ : surface
 $\partial\Sigma$: boundary curve
 Ω : volume
 $\partial\Omega$: boundary surface

Learning objectives

Students will be able to:

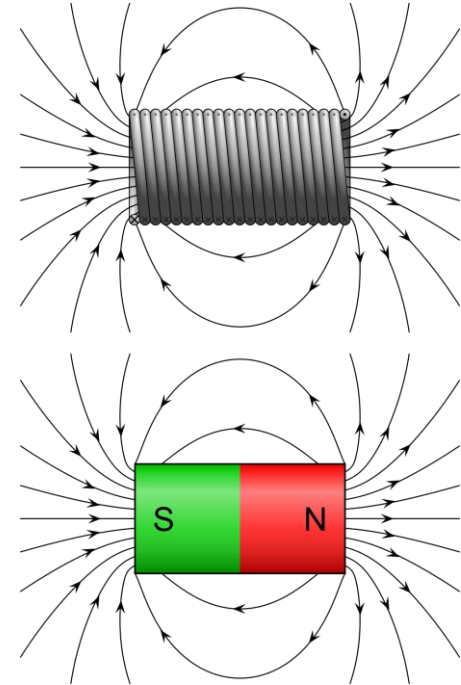
- Summarize important **properties of magnetic fields**.
- Calculate the **magnetic force** on a **moving charge**.
- Describe the difference between **magnetic field intensity** and **flux density**.
- Describe the concept of **magnetic flux**.
- Name **Gauss' law of magnetism** and derive why there are **no magnetic monopoles**.
- Explain the **analogies and differences** between **electric** and **magnetic fields**.

3.2 Basic concepts of magnetic fields

3.2 Basic concepts of magnetic fields

Properties of magnetic fields

- Vector field $\vec{B}$ describing the **magnetic influence** on
 - ⌈ **moving** electric **charges**.
 - ⌈ electric **currents**.
 - ⌈ **magnetic** materials.
- Magnetic fields $\vec{H}$ can be generated by ...
 - ⌈ ... **moving electric charge** / electric currents.
 - ⌈ In general: law of Biot-Savart.
 - ⌈ High symmetry: **Ampère's circuital law**.
 - ⌈ ... intrinsic magnetic moments in **permanent magnets**.
- Outside a bar magnet, field lines run **from the north pole to the south pole**.
- Changing magnetic fields **induce a voltage**.



*A solenoid (**top**) with electric current running through it has the same properties as a bar magnet (**bottom**).*

3.2 Basic concepts of magnetic fields

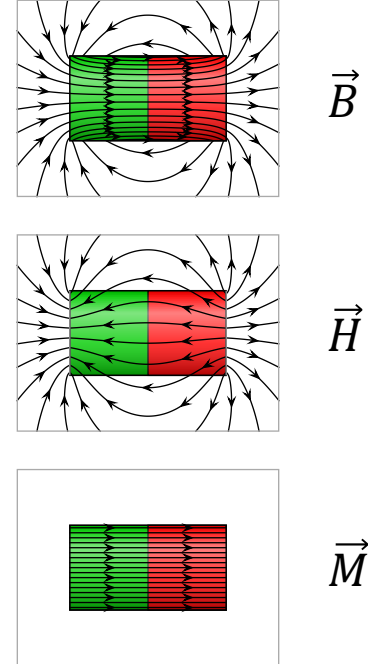
Relation of $\vec{B}$, $\vec{H}$ and $\vec{M}$

- **Magnetic flux density** $\vec{B}$, $[B] = \text{T} = \frac{\text{Vs}}{\text{m}^2} = \frac{\text{Wb}}{\text{m}^2}$
 - ⌋ **Effect (force)** on a moving **charge** (or magnet) analogue to the electric field intensity $\vec{E}$.
 - ⌋ But: mathematical symmetry to **electric flux density** $\vec{D}$.
- **Magnetic field intensity** $\vec{H}$, $[H] = \frac{\text{A}}{\text{m}}$
 - ⌋ **Excitation of a magnetic field** generated by **electric currents** analogue to electric flux density $\vec{D}$.
 - ⌋ But: mathematical symmetry to **the electric field intensity** $\vec{E}$
- **Magnetization** $\vec{M}$, $[M] = \frac{\text{A}}{\text{m}}$
 - ⌋ Density of **magnetic dipole moment** inside the material.
 - ⌋ Analogue to the electric polarization.
- In linear, homogeneous media the relation between $\vec{B}$ and $\vec{H}$ is.

$$\vec{B} = \mu_0 (\vec{H} + \vec{M}) = \mu_0 \vec{H} + \underbrace{\mu_0 \chi_m \vec{H}}_{\text{magnetization}} = \mu_0 \mu_r \vec{H}$$

 - ⌋ With the vacuum permeability $\mu_0 = \frac{1}{\epsilon_0 c^2} = 4 \pi \cdot 10^{-7} \frac{\text{Vs}}{\text{A m}}$, the relative permeability μ_r and the magnetic susceptibility χ_m .

Similar but opposite principle, due to the different origin of electric fields and magnetic fields.



$\vec{B}$, $\vec{H}$ and $\vec{M}$ inside and outside a cylindrical bar magnet. Arrows go **from north pole (red) to south pole (green)**.

3. Magnetic fields

Mathematical nature of static electric and magnetic fields in a linear medium

Electric field

- Derived from a **scalar potential** φ
- Electric field intensity via **grad**: $\vec{E} = -\vec{\nabla} \varphi$
- Electric fields do **not rotate**

$$\vec{\nabla} \times \vec{E} = \vec{\nabla} \times (\vec{\nabla} \phi) = 0$$

- Electric fields **diverge from a charge density** ρ

$$\vec{\nabla} \cdot \vec{E} = -\vec{\nabla} \cdot (\vec{\nabla} \phi) = \frac{\rho}{\epsilon}$$

- **Outside** of a charge density the field does **not diverge**!

- Relation of **flux density** $\vec{D}$ and **field intensity** $\vec{E}$

$$\vec{D} = \epsilon \vec{E}$$



Depiction of a positive (**left**) and negative (**right**) charge including electric flux density $\vec{D}$.

Magnetic field

- Derived from a **vector potential** $\vec{A}$
- Magnetic flux density via **curl**: $\vec{B} = \vec{\nabla} \times \vec{A}$
- Magnetic fields are **source-free**

$$\vec{\nabla} \cdot \vec{B} = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

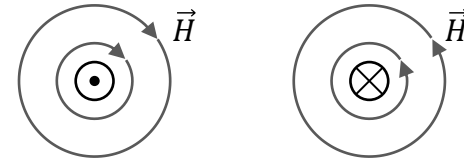
- Magnetic fields **rotate around a current density** $\vec{j}$

$$\vec{\nabla} \times \vec{B} = \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \mu \vec{j}$$

- **Outside** of a current density the field does **not rotate**!

- Relation of **flux density** $\vec{B}$ and **field intensity** $\vec{H}$

$$\vec{B} = \mu \vec{H}$$



Depiction of a current entering (**left**) and leaving (**right**) the plane including magnetic field intensity.

3. Magnetic fields

Analogy between magnetic and electrical fields



Here you see the 'similar but different' principle in perfection! 😊

Electric		
Name	Symbol	Unit
Electric field intensity (effect, depends on medium)	$\vec{E}$	$\frac{\text{V}}{\text{m}}$
Electric flux density (source, depends on charge)	$\vec{D}$	$\frac{\text{A s}}{\text{m}^2}$
Electric flux	$\Phi = \ \vec{D}\ \ \vec{A}\ $	A s
Relation between D and E	$\vec{D} = \varepsilon \vec{E}$	
Permittivity	$\varepsilon = \varepsilon_r \varepsilon_0$	$\frac{\text{A s}}{\text{V m}}$
Capacitance	$C = \frac{Q}{U}$	$\text{F} = \frac{\text{A s}}{\text{V}}$
Voltage (electromotive force EMF)	$U = \int \vec{E} \cdot d\vec{l}$	V
Maxwell's 1st (Gauss)	$\oiint_{\partial\Omega} \vec{D} \cdot d\vec{A} = \iiint_{\Omega} \vec{\nabla} \cdot \vec{D} dV = Q_{\text{encl.}}$	
Maxwell's 2nd (Stoke's law)	$\oint_{\partial\Sigma} \vec{E} \cdot d\vec{l} = \iint_{\Sigma} \vec{\nabla} \times \vec{E} \cdot d\vec{A} = 0$	

Magnetic		
Name	Symbol	Unit
Magnetic field intensity (source, depends on current)	$\vec{H}$	$\frac{\text{A}}{\text{m}}$
Magnetic flux density (effect, depends on medium)	$\vec{B}$	$\text{T} = \frac{\text{V s}}{\text{m}^2}$
Magnetic flux	$\Phi = \ \vec{B}\ \ \vec{A}\ $	Wb = V s
Relation between B and H	$\vec{B} = \mu \vec{H}$	
Permeability	$\mu = \mu_r \mu_0$	$\frac{\text{V s}}{\text{A m}}$
(Self) inductance	$L = N \frac{\Phi}{i}$	$\text{H} = \frac{\text{V s}}{\text{A}}$
'Magnetic voltage' (magnetomotive force MMF)	$V_m = \int \vec{H} \cdot d\vec{l}$	A
Maxwell's 3rd (Gauss)	$\oiint_{\partial\Omega} \vec{B} \cdot d\vec{A} = \iiint_{\Omega} \vec{\nabla} \cdot \vec{B} dV = 0$	
Maxwell's 4th (Ampère's circuital law)	$\oint_{\partial\Sigma} \vec{H} \cdot d\vec{l} = \iint_{\Sigma} \vec{\nabla} \times \vec{H} \cdot d\vec{A} = I$	

3.2 Basic concepts of magnetic fields

More definitions of $\vec{B}$ and $\vec{H}$

Publication	Definition of magnetic field	Definition of magnetic field strength $\vec{H}$	Definition of magnetic flux density $\vec{B}$
R. Feynman, R. Leighton, M. Sands The Feynman Lectures on Physics	<i>First, we must extend, somewhat, our ideas of the electric and magnetic vectors, $\vec{E}$ and $\vec{B}$. We have defined them in terms of the forces that are felt by a charge. We wish now to speak of electric and magnetic fields at a point even when there is no charge present. We are saying, in effect, that since there are forces “acting on” the charge, there is still “something” there when the charge is removed.</i>	<i>[...] Most people who use the mks units have chosen to use a different definition of $\vec{H}$. Calling their field $\vec{H}'$ (of course, they still call it $\vec{H}$ without the prime), it is defined by</i> $\vec{H}' = \epsilon_0 c^2 \vec{B} - \vec{M}$ <i>Also, they usually write $\epsilon_0 c^2$ as a new number $\frac{1}{\mu_0}$</i>	<i>We can write the force $\vec{F}$ on a charge q moving with a velocity $\vec{v}$ as</i> $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ <i>We call $\vec{E}$ the electric field and $\vec{B}$ the magnetic field at the location of the charge.</i>
David C. Jiles Introduction to Magnetism and Magnetic Materials	<i>One of the most fundamental ideas in magnetism is the concept of the magnetic field. When a field is generated in a volume of space it means that there is a change of energy of that volume, and furthermore that there is an energy gradient so that a force is produced which can be detected by the acceleration of an electric charge moving in the field, by the force on a current-carrying conductor, by the torque on a magnetic dipole such as a bar magnet or even by a reorientation of spins of electrons within certain types of atoms.</i>	<i>There are a number of ways in which the magnetic field strength $\vec{H}$ can be defined. In accordance with the ideas developed here we wish to emphasize the connection between the magnetic field $\vec{H}$ and the generating electric current. [...] The simplest definition is as follows. The Ampère per meter is the field strength produced by an infinitely long solenoid containing n turns per metre of coil and carrying a current of $\frac{1}{n}$ Amperes.</i>	<i>When a magnetic field $\vec{H}$ has been generated in a medium by a current, in accordance with Ampère's law, the response of the medium is its magnetic induction $\vec{B}$, also sometimes called the flux density.</i>
Magnetic field, Encyclopaedia Britannica	<i>Magnetic field, region in the neighbourhood of a magnetic, electric current, or changing electric field, in which magnetic forces are observable.</i>	<i>The magnetic field $\vec{H}$ might be thought of as the magnetic field produced by the flow of current in wires [...]</i>	<i>[...] the magnetic field $\vec{B}$ [might be thought of] as the total magnetic field including also the contribution made by the magnetic properties of the materials in the field.</i>

3.2 Basic concepts of magnetic fields

Magnetic force

- Observation when a **magnet** is placed next to a **cathode-ray tube**:

- ▮ The direction of the **force** $\vec{F}$ is **perpendicular** both to the **velocity** $\vec{v}$ and the **magnetic flux density** $\vec{B}$:

$$\vec{F} \perp \vec{B} \quad \& \quad \vec{F} \perp \vec{v}$$

- ▮ The **magnitude** $\|\vec{F}\|$ is **proportional** to the **angle** θ between $\vec{v}$ and $\vec{B}$:

$$\|\vec{F}\| \propto \sin \theta$$

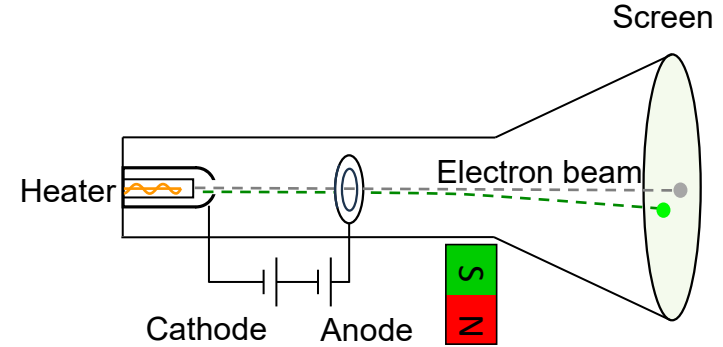
- This behaviour can be described by the **cross product** of the **velocity** $\vec{v}$ with the **magnetic flux density** $\vec{B}$ and is called **Lorentz Force**.

- ▮ Moving charge with velocity

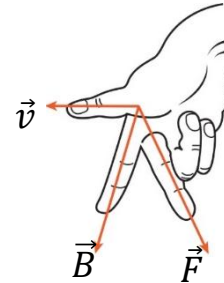
$$\vec{F} = Q \vec{v} \times \vec{B} \quad \text{and} \quad \|\vec{F}\| = |Q| \|\vec{v}\| \|\vec{B}\| \sin \theta$$

- ▮ Current carrying wire

$$\vec{F} = I \vec{l} \times \vec{B} \quad \text{and} \quad \|\vec{F}\| = |I| \|\vec{l}\| \|\vec{B}\| \sin \theta$$



Experiment with a cathode-ray tube to demonstrate the deflection of an electron beam.



For electrons, the force direction is opposite to the right-hand-rule result for a positive charge.

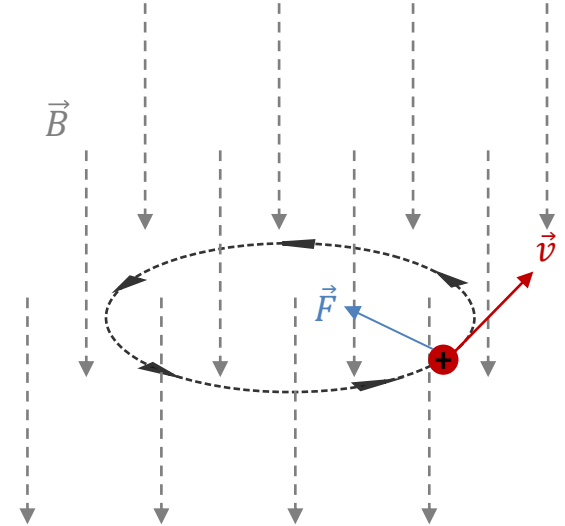
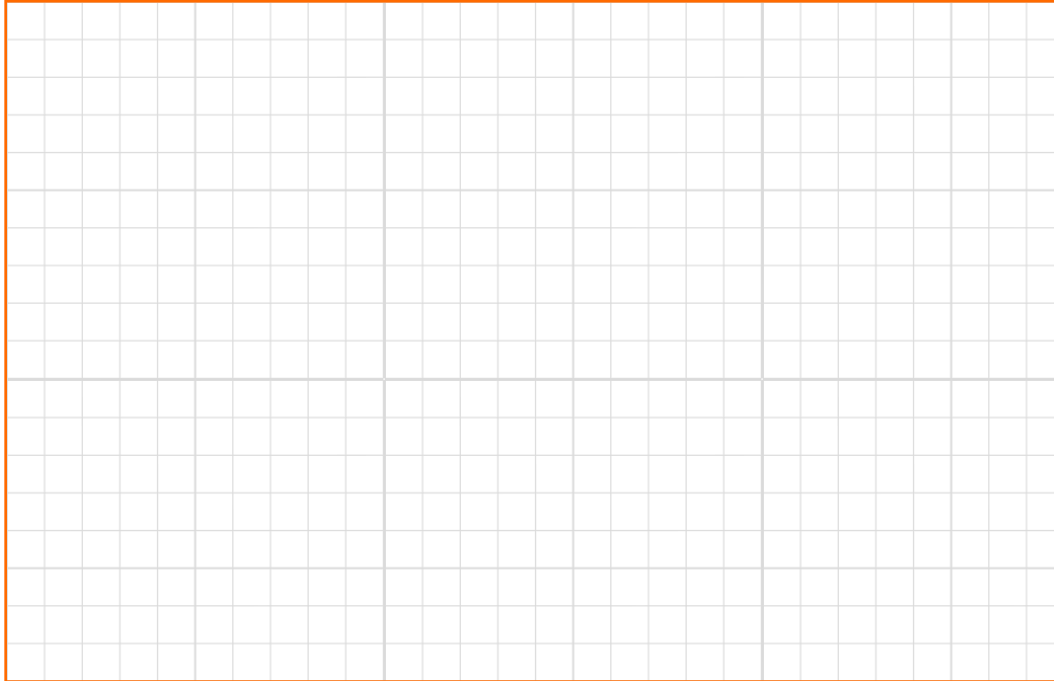
Right-hand rule to find the direction of the Lorentz-Force.

3.2 Basic concepts of magnetic fields

Exercise: Trajectory of a charge in a magnetic field

EXAM	Typical
•	Exam
•	Question
•	
•	

A positively charged particle is moving on a trajectory perpendicular to a magnetic flux density $\vec{B}$. Describe the shape of the trajectory. What would happen if the trajectory would have a component in the direction of $\vec{B}$?

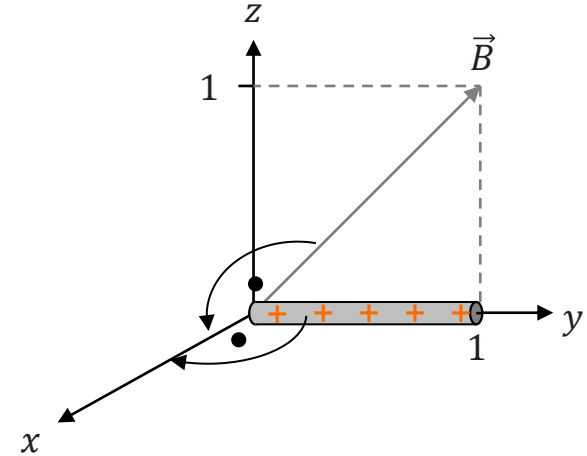


3.2 Basic concepts of magnetic fields

Exercise: Magnetic force on a current carrying wire

EXAM	Typical
• —	Exam
• —	Question
• —	

Derive the magnetic force $\vec{F}$ on a straight wire of length l carrying current I from the Lorentz force. Then evaluate it for $\vec{B} = (0,1,1)$ T, $\vec{l} = (0,1,0)$ m, $I = 1$ A.



3.2 Basic concepts of magnetic fields

Surface orientation and sign of flux

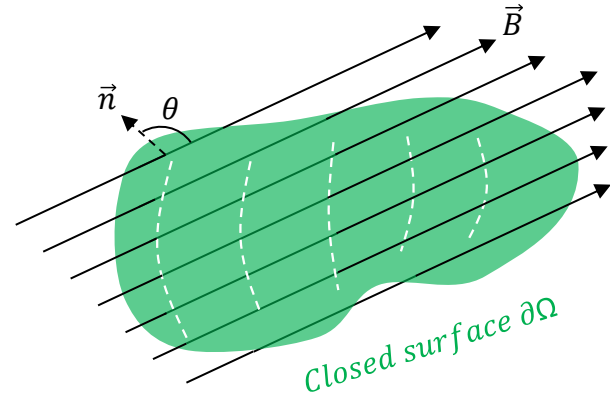
- Flux uses the area vector $d\vec{A} = \vec{n} dA$, normal to the surface.
- Therefore, flux depends on the angle between $\vec{B}$ and the surface normal $\vec{n}$:

$$\Phi = \oiint_{\partial\Omega} \vec{B} \cdot d\vec{A}$$

- Uniform field on flat surface:

$$\Phi = \|\vec{B}\| \|\vec{A}\| \cos \theta$$

- ⌈ Field parallel to surface normal: $\theta = 0^\circ$: maximum flux
- ⌈ Field parallel but opposite to surface normal = 180° : maximum negative flux
- ⌈ Field perpendicular to surface normal: $\theta = 90^\circ$: zero flux



3.2 Basic concepts of magnetic fields

Magnetic flux

- Magnetic flux ϕ is the **surface integral of the magnetic flux density** through a surface,

$$\Phi = \underbrace{\oint_{\partial\Omega} \vec{B} \cdot d\vec{A}}_{\text{field lines through } \partial\Omega} = \underbrace{\iiint_{\Omega} \vec{\nabla} \cdot \vec{B} dV}_{\text{divergence in } \Omega} \quad [\Phi] = \text{T m}^2 = \text{Vs} = \text{Wb}$$

- **Physical intuition:** Number of field lines crossing a surface.

- For a uniform field and a flat surface

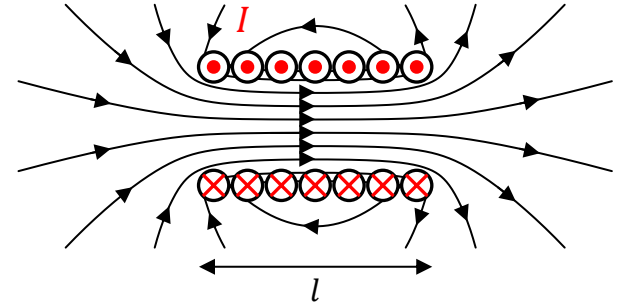
$$\Phi = \|\vec{B}\| \|\vec{A}\| \cos \theta$$

- If B is parallel to the surface normal:

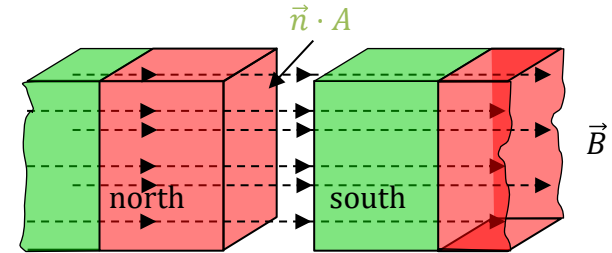
$$\Phi = \|\vec{B}\| \|\vec{A}\|$$

- *Note: In a magnetic circuit of varying diameter, the flux density will change, the total flux is kept constant!*

Current into the plane: $\otimes$
Current out of the plane: $\odot$



Schematic of solenoid of length l including flux illustrated by the 'number of field lines' inside.



The magnetic flux Φ between two bar magnets is equivalent to the magnitude of the flux density B and the area A .

3.2 Basic concepts of magnetic fields

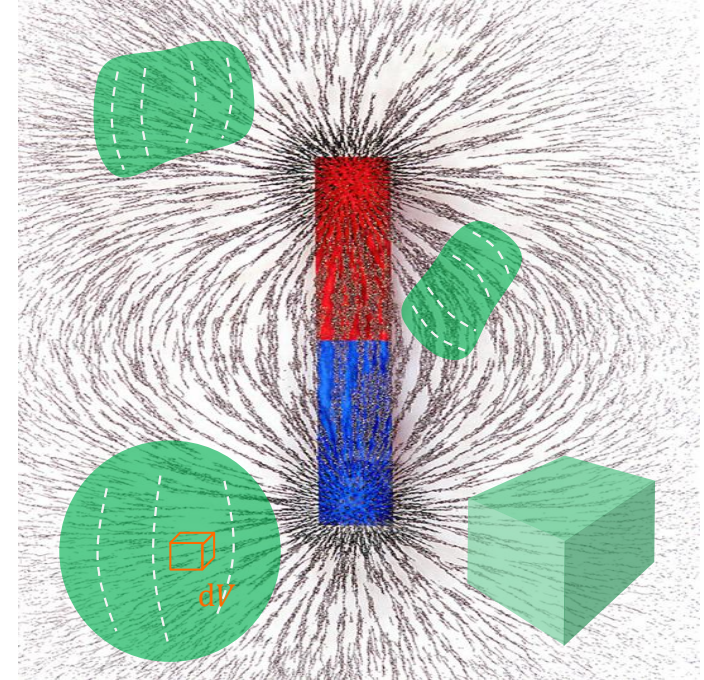
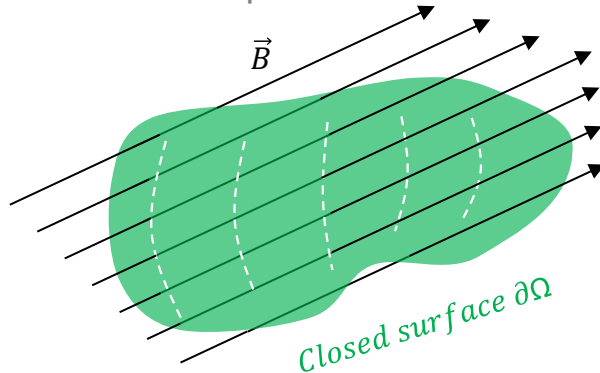
Maxwell's 3rd equation

- Magnetic field lines form closed loops
- Therefore, every field line entering a closed surface must also leave it.

$$\Phi = \underbrace{\oint_{\partial\Omega} \vec{B} \cdot d\vec{A}}_{\text{"field lines through } \partial\Omega"} = \underbrace{\iiint_{\Omega} \vec{\nabla} \cdot \vec{B} dV}_{\text{divergence in } \Omega} = 0$$

*"The magnetic flux Φ through a closed surface $\partial\Omega$ is zero.
There are no free magnetic poles!"*

— Maxwell's 3rd equation / Gauss' law of magnetics



The magnetic flux through a closed surface $\partial\Omega$ is zero.

Summary

- Magnetic fields **exert influence** on **moving charges**: free charges, current carrying wires and magnetic material
- The **force** is acting **perpendicular** to the **direction of movement** and the **field**. Use the right-hand rule to find the direction.
- The **magnetic field** $\vec{H}$ represents the source of the magnetic field generated by an electric current.
- The **magnetic flux density** $\vec{B}$ describes the effect on moving charges.
- If the magnetic field is parallel to the surface normal, the flux equals $\|\vec{B}\| \|\vec{A}\|$.
- All **surfaces without boundary** are **closed**: Boxes, spheres, etc.
- Maxwell's 3rd equation: "The magnetic flux through a closed surface is zero. There are no free magnetic poles."

3.2 Basic concepts of magnetic fields

Learning objectives

Students will be able to:

- Explain the nature of **magnetic fields around current carrying wires**.
- Describe the **shape** of the magnetic field around **different wire layouts**.
- Explain the **concept** and **formalism** of the **magnetic voltage** and the **magnetomotive force MMF**.
- Apply **Ampère's circuital law** to calculate the magnetic field for basic setups.

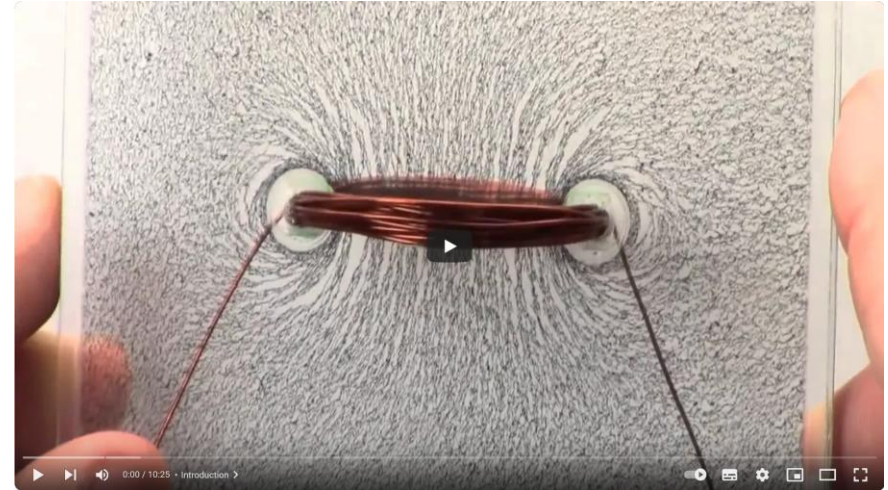
3.3 Stationary magnetic fields

3.3 Stationary magnetic fields

Analyse the shapes around different setups

A **current carrying** wire generates a magnetic field

- ⌈ **Concentric circles** around the wire.
- ⌈ The direction of the magnetic field is **perpendicular** to the **current** direction.
- ⌈ The **magnitude** of the field depends on the **current I** and the **distance r** from the wire.
- ⌈ For the same geometry and current per turn, **increasing the number of turns strengthens the field**.



3.3 Magnetic field around a current carrying wire

Magnetic field intensity generated by electric currents

A **steady current** generates a **magnetic field** whose strength and direction depend on the **current**, the **wire geometry**, and the **distance** and **direction** from the point of observation.

– Law of Biot-Savart

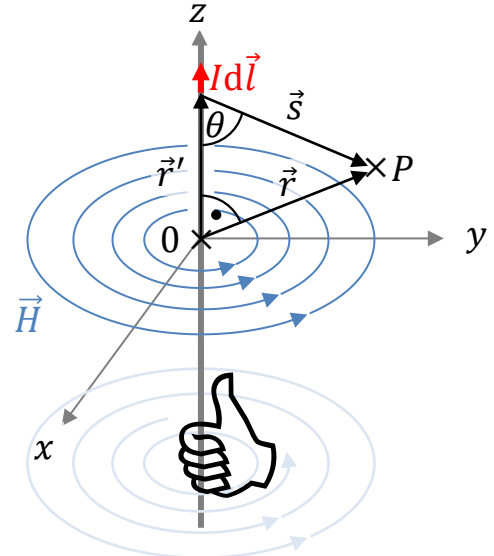
- The magnetic field intensity is given by the Biot–Savart law

$$\vec{H}(\vec{r}) = \frac{I}{4\pi} \int_{\text{wire}} \frac{d\vec{l} \times \vec{s}}{\|\vec{s}\|^3}$$

- ▮ Where $\vec{r}$ is the observation point, $\vec{r}'$ the source point on the wire, $d\vec{l}$ the local tangent element at $\vec{r}'$, and $\vec{s} = \vec{r} - \vec{r}'$ the vector from source to observation point.
- If only the magnitude is of interest, we use

$$\|\vec{H}(\vec{r})\| = \frac{I}{4\pi} \int_{\text{wire}} \frac{\|d\vec{l} \times \vec{s}\|}{\|\vec{s}\|^3} = \frac{I}{4\pi} \int_{\text{wire}} \frac{\sin \theta dl}{\|\vec{s}\|^2}$$

- ▮ Where θ is the angle between $d\vec{l}$ and $\vec{s}$.



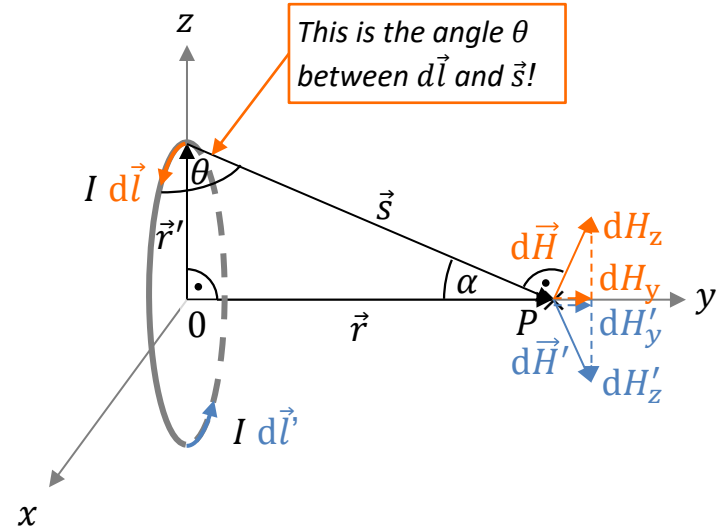
Right-hand grip rule: thumb in current direction, curled fingers show the field direction.

Magnetic field $\vec{B}$ around a current carrying wire. Each element $I d\vec{l}$ contributes to $\vec{H}$ at P-

3.3 Magnetic field around a current carrying wire

Example: Magnetic field density of a loop of wire 1/2

Apply Biot–Savart's law to calculate $\|\vec{H}(\vec{r})\|$ and $\vec{H}(\vec{r})$ at a point P on the symmetry axis of a circular loop of radius $\|\vec{r}'\|$ carrying a constant current I .

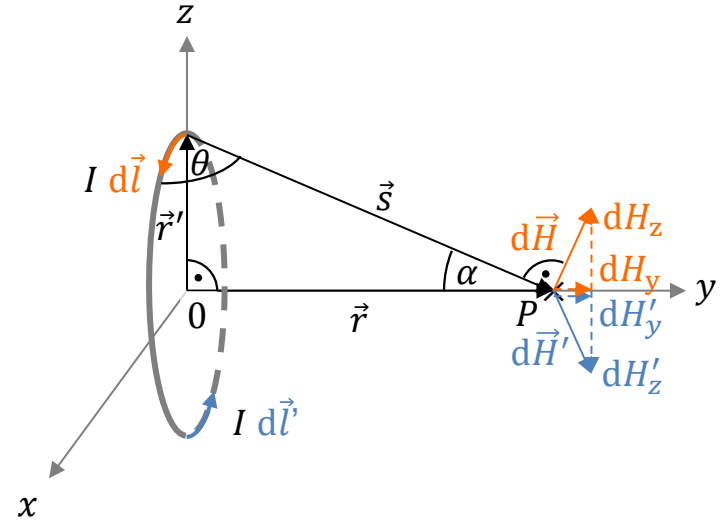


A current carrying wire, opposite wire elements cancel x and z component of $\vec{H}$.

3.3 Magnetic field around a current carrying wire

Example: Magnetic field density of a loop of wire 2/2

For which distance $\|\vec{r}\|$ of the ring centre in direction of y is $\|\vec{H}(\vec{r})\|$ maximised and minimized, respectively. Calculate $\|\vec{H}(\vec{r})\|$ at this point. What is the dependency of $\|\vec{r}\|$ for large distances?



A current carrying wire, opposite wire elements cancel x and z component of $\vec{H}$.

3.3 Magnetic field around a current carrying wire

Ampère's circuital law

- Choose an **imaginary closed path** $\partial\Sigma$ around the current
- The **circulation** Γ , also called magnetomotive force MMF, of the magnetic field intensity of the magnetic field intensity along this closed path equals the enclosed current.

$$\Gamma = \oint_{\partial\Sigma} \vec{H} \cdot d\vec{l} = I_{\text{enc}}$$

“Electric current is the source for magnetic fields.”

– Maxwell's 3rd / Ampère's circuital law

- For N turns carrying current I :

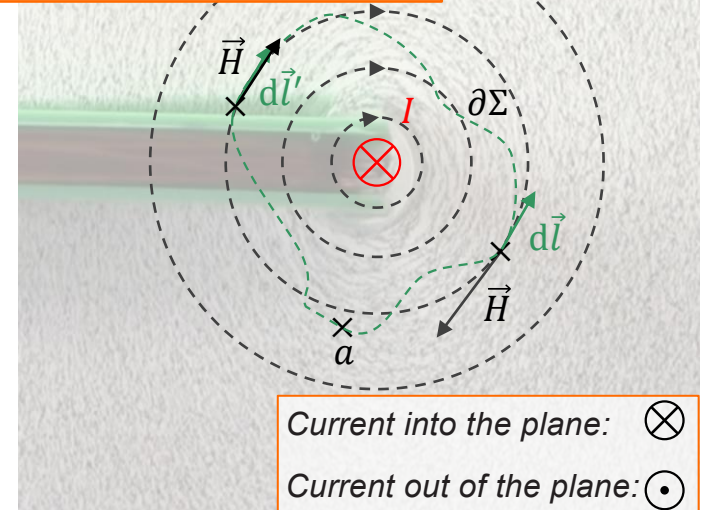
$$\oint_{\partial\Sigma} \vec{H} \cdot d\vec{l} = N I$$

- If $\vec{H}$ is **tangential** to the path along it: $\oint_{\partial\Sigma} \|\vec{H}\| dl = N I$
- If $\vec{H}$ is **tangential** to the path along it **and constant** :

$$\|\vec{H}\| \|\vec{l}\| = N I$$



Because it also equals
 $\oint_{\partial\Sigma} \vec{H} \cdot d\vec{l} = \iint_{\Sigma} \vec{\nabla} \times \vec{H} \cdot d\vec{A}$
magnetic fields rotate
round current densities.



The circulation Γ depends on the path around the current carrying wire. Clockwise direction: $\Gamma > 0$; Counter-clockwise direction: $\Gamma < 0$.

3.3 Magnetic field around a current carrying wire

Ampère's circuital law for common elements

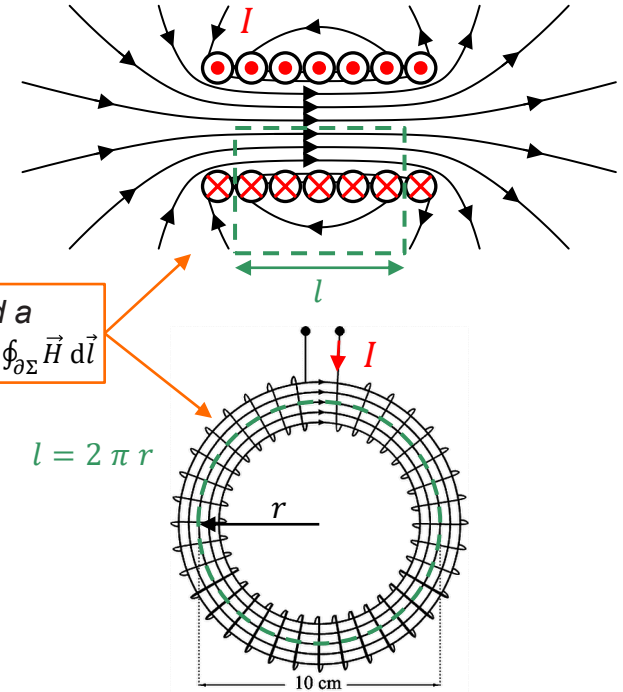
- **Solenoid** of with N turns per length l
 - the **flux is confined inside** in form of a **homogenous field parallel to the path**
 - The field at the **sides** is **perpendicular** to the path.
 - The field **outside** can be **neglected**.
 - The field inside calculates as

$$\|\vec{H}\| = \frac{N I}{l}$$

- **Toroidal coil**
 - Can be considered a **bent solenoid** of length $l = 2 \pi r$

$$\|\vec{H}\| = \frac{N I}{2 \pi r}$$

- *Note: The magnetic field $\vec{H}$ is independent of the core material. If the volume inside is filled with magnetic material, the magnetic flux density will increase according to $\vec{B} = \mu \vec{H}$.*



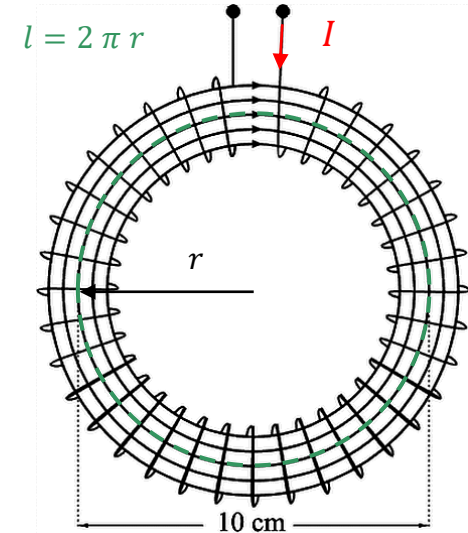
Solenoid (**top**) and toroidal coil (**bottom**) including path in blue.

3.3 Stationary magnetic fields

Example: Magnetic field inside a toroidal coil

EXAM	Typical
•	Exam
•	Question
•	
•	

Calculate the magnetic field $\|\vec{H}\|$ and the magnetic flux density $\|\vec{B}\|$ inside a toroidal coil with $N = 200$ windings, a diameter $d = 10$ cm and a current $I = 0.94$ A filled with a) air and b) with iron with $\mu_r = 2000$.



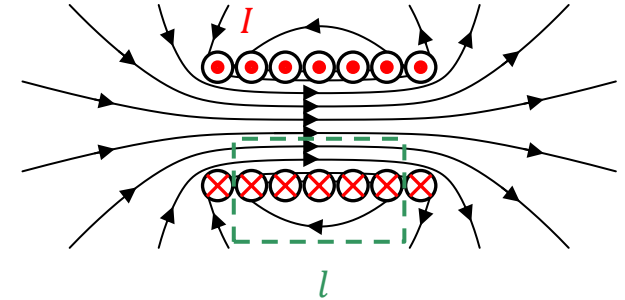
Toroidal coil including the closed path according to Ampère's circuital law.

3.3 Magnetic field around a current carrying wire

Homework: Magnetic field inside a solenoid

EXAM	Typical
•	Exam
•	Question
•	

Calculate the magnetic field $\|\vec{H}\|$ inside an air filled solenoid when a current of $I = 0.2 \text{ A}$ is driven through the wires of a wire density of 500 cm^{-1} using Ampère's circuital law. Does the length of your path matter?



Solenoid including the closed path according to Ampère's circuital law.

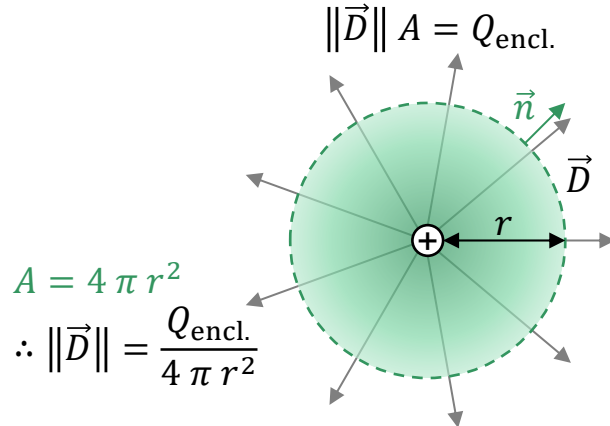
3. Magnetic fields

Gauss' law and Ampère's circuital law

Maxwell's first equation / Gauss' law:

*"The **flux** through a closed surface is **independent of the surface area** and shape and equal to the **charge enclosed by the surface**."*

$\vec{D}$ constant on the **Gaussian surface** and $\vec{D} \parallel \vec{n}$, then:

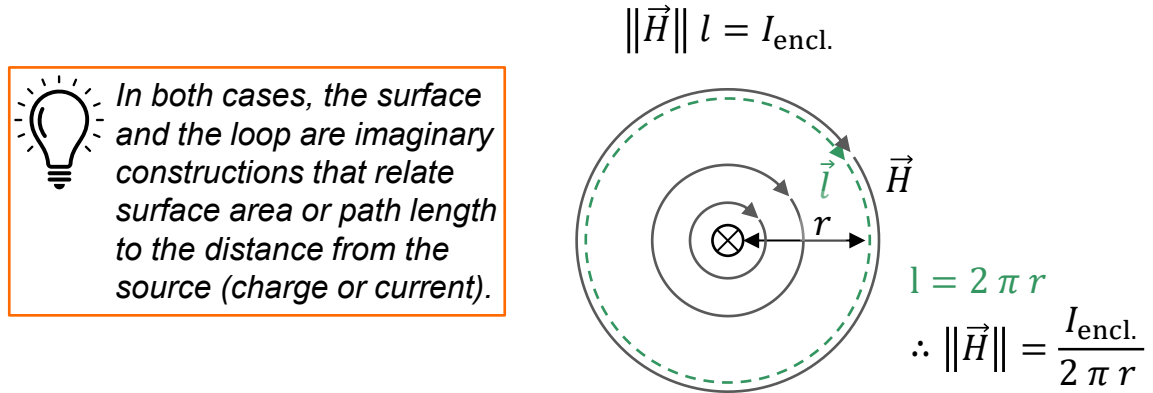


Application of Gauss' law around to calculate $\|\vec{D}\|$ at a distance r around a current carrying wire.

Maxwell's 3rd / Ampère's circuital law:

„Electric currents create steady magnetic field loops around them“

$\vec{H}$ constant along the **loop around** and $\vec{H} \parallel \vec{l}$, then:



Application of Ampère's circuital law to calculate $\|\vec{H}\|$ at a distance r around a current carrying wire.

Summary

- The **magnetic field** around a **current carrying wire** form **concentric circles perpendicular to the wire**.
- The **magnitude** of the magnetic field **depends on the total current** $N I$ and the **distance**.
- In general, the magnetic flux density is calculated via the law of **Biot-Savart**.
- More practical is **Ampère's circuital law**, where circulation around a closed path in a **magnetic field intensity is equal to the current** confined by this path.
- If the **path is colinear to a homogeneous magnetic field**, **Ampère's circuital law is simply**

$$\|\vec{H}\| \|\vec{l}\| = N I$$

- Ampère's circuital law around a closed path is analogous to Gauss' law for electric flux through a closed surface.

3.3 Stationary magnetic fields

Learning objectives

Students will be able to:

- Explain the **cause for magnetization** of matter.
- Relate the **magnetic flux density** to the **magnetic field intensity** using **magnetization curves**.
- Extract the **relative permeability** from magnetization curves.
- Explain magnetic **hysteresis**.
- Select the right **core material** for your application.

3.4 Magnetization of matter

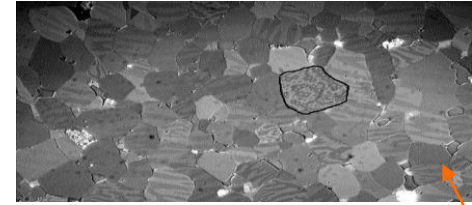
3.4 Magnetization of matter

Ferromagnetic materials

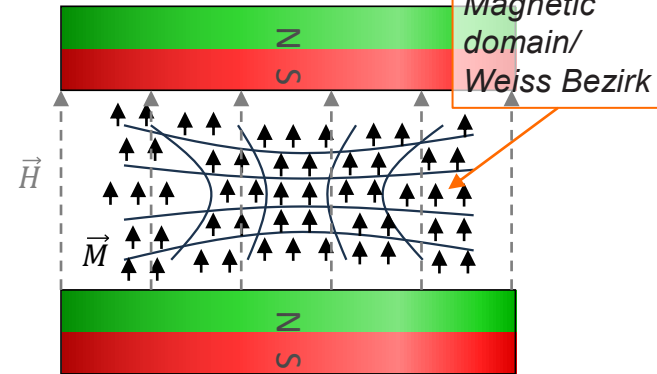
- Magnetization $\vec{M}$ comes from **bound microscopic currents** in the material.
- Magnetic moments align easily in a small applied field.
 - ⌈ Domains are **locally saturated** $\|\vec{M}\| = \|\vec{M}_{\text{sat}}\|$.
 - ⌈ **High** magnetic **permeability** $\mu_r \gg 1$.
 - ⌈ Paramagnetic above the **critical temperature** T_C .
 - ⌈ **High coercivity** (resistance to demagnetization).
- When all domains are aligned, the material is **saturated** and further magnetization is no longer possible
- The magnetic flux density is:
- For a linear, homogeneous material, $\vec{M} = \chi_m \vec{H}$:

$$\vec{B} = \mu_0(1 + \chi_m)\vec{H} = \mu_0 \mu_r \vec{H}$$

Here, χ_m is the magnetic susceptibility and μ_r the relative permeability



Microcrystalline grains within a piece of Nd₂Fe₁₄B with magnetic domains made visible with a Kerr microscope.



All domains inside a ferromagnet are aligned under the influence of an external mag. field.

3.4 Magnetization of matter

Magnetization curves

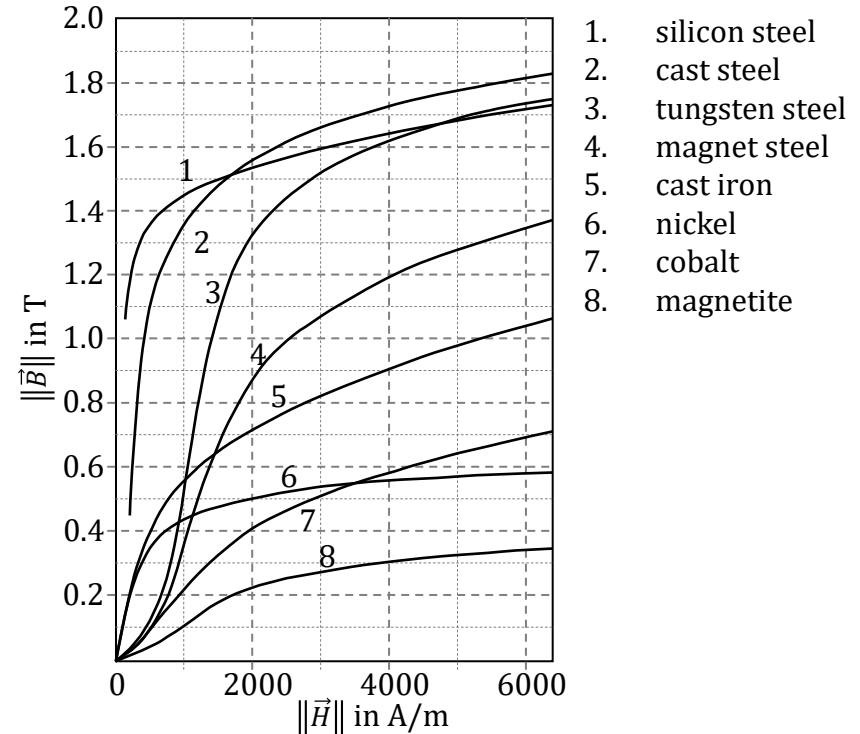
- The alignment of **spins** in a ferromagnet with an external **magnetic field** is called **magnetization**.
- Magnetization curves relate the external magnetic field to the magnetic flux density via μ_r :

$$\|\vec{B}\| = \mu_0 \mu_r \|\vec{H}\|$$

- This relation is nonlinear. A field beyond **saturation** H_s does not increase magnetization (all magnetic domains are aligned).
- The magnetization influences the application
 - ⌈ **Higher magnetization** enables **higher magnetic flux** for a given **field** $\|\vec{H}\|$.
 - ⌈ **Higher magnetization** reduces the required current (and thus energy) for a target $\|\vec{B}\|$.



Other than a **dielectric damping** the electric field, most **magnetic materials enhance** the magnetic field.



Magnetization curves of different magnetic materials.

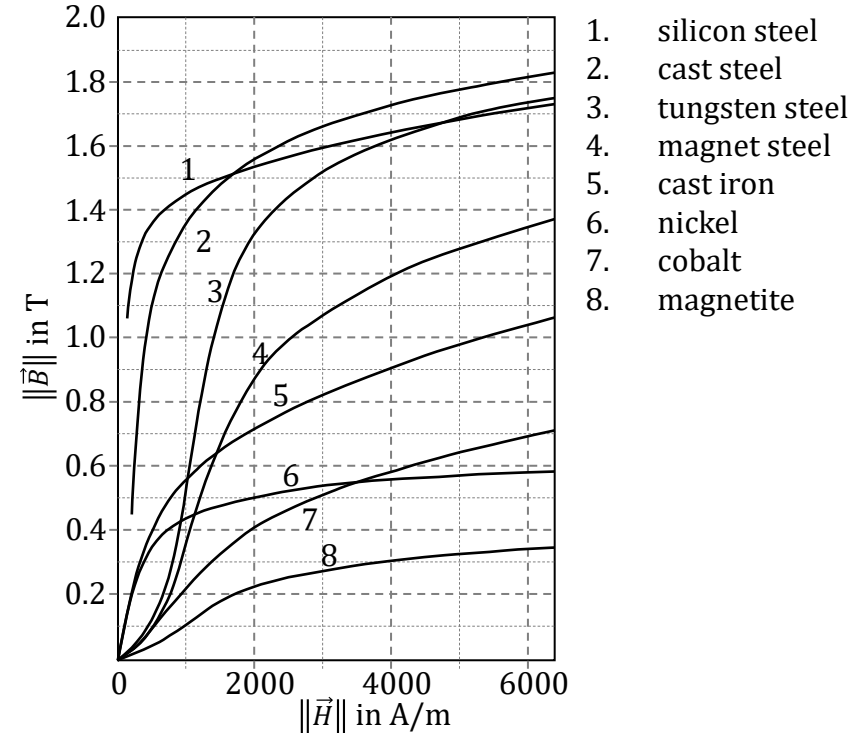
3.4 Magnetization of matter

Example: Magnetization curves

Find the magnetic flux density and the relative permeability of silicon steel and magnetite for a magnetic field $\|\vec{H}\| = 1000 \text{ A/m}$.



EXAM	Typical
•	Exam
•	Question



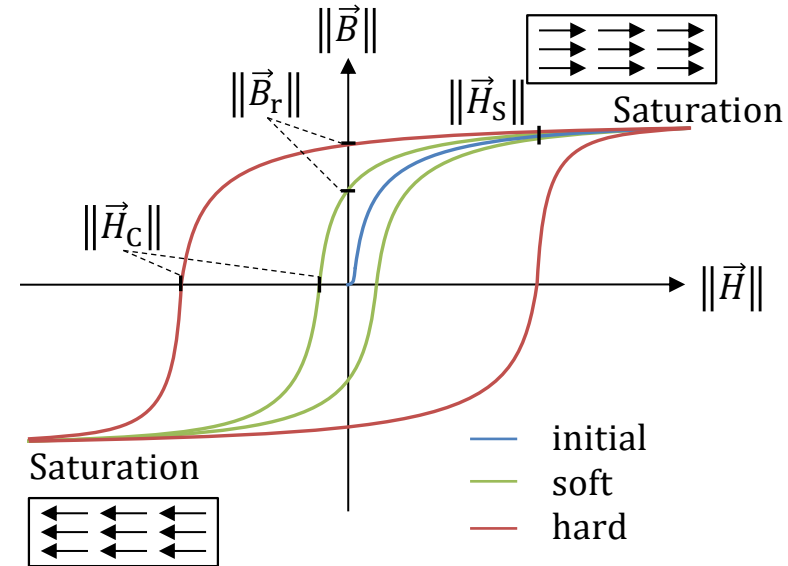
Magnetization curve of selected core materials.

3.4 Magnetization of matter

Magnetic hysteresis

- When a ferromagnetic material is magnetized and then demagnetized, B or M does not follow the same path back. This history dependence is called **hysteresis**.
- **Remanence**: remaining magnetization at $H = 0$.
- **Coercive field** H_C : reverse field needed to demagnetize.

Soft magnetic materials	Hard magnetic materials
Small hysteresis loop, small H_C	Wide hysteresis loop, large H_C
Easy to magnetize and demagnetize	Difficult to demagnetize
Used for transformer and electromagnet cores	Used for permanent magnets, e.g. motors and loudspeakers



Hysteresis curve of “soft” and “hard” ferromagnetic material with initial magnetization curve.

3.4 Magnetization of matter

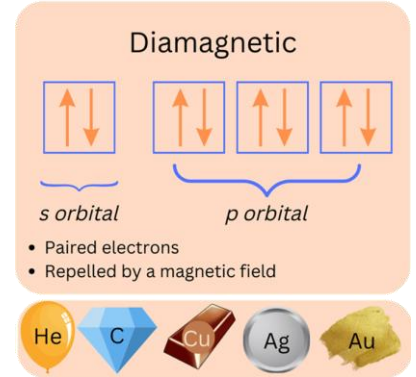
Diamagnetic and Paramagnetic Materials

Diamagnetic

- Electron moments nearly cancel out.
- External field induces a small magnetization opposite to the field.

$$\vec{M} < 0 \text{ (small!)}, \quad \mu_r < 1$$

- Weakly repelled by a magnetic field.
- *Example:* copper with $\mu_r = 0.999994$

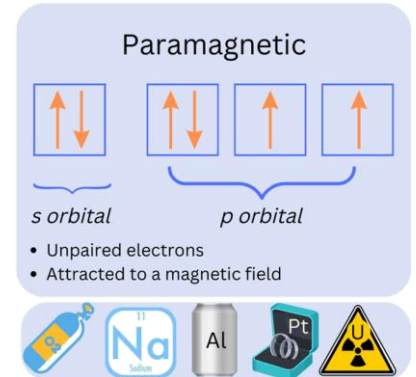


Paramagnetic

- Unpaired electrons provide magnetic moments.
 - Without external field, thermal agitation randomizes the moments.
- External field induces a small magnetization along the field.

$$\vec{M} > 0 \text{ (small!)}, \quad \mu_r > 1$$

- Weakly attracted by a magnetic field.
- *Example:* aluminium with $\mu_r = 1.000022$



Summary

- Magnetization is the alignment of microscopic magnetic dipoles (spinning electrons) inside matter.
- Ferromagnetic materials can be strongly magnetized, show nonlinear B - H behaviour, and may exhibit saturation and hysteresis
- Diamagnetic and paramagnetic materials respond only weakly to an external magnetic field and are often approximated as vacuum in engineering calculations
- Magnetization curves relate the applied field $\|\vec{H}\|$ to the resulting flux density $\|\vec{B}\|$ and allow the relative permeability μ_r to be estimated.
- Soft magnetic materials are suitable for cores
- Hard magnetic materials are suitable for permanent magnets

3.4 Magnetization of matter

Learning objective

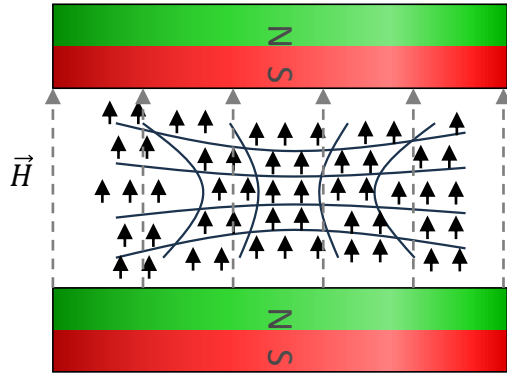
Students will be able to:

- Explain the concept of magnetic circuits.
- Apply the concepts of magnetic flux and magnetomotive force to calculate magnetic circuits.
- *Note: Throughout this chapter we use the practical convention $B := \|\vec{B}\|$ and $H := \|\vec{H}\|$, i.e. B and H denote the magnitudes of the corresponding vector fields.*

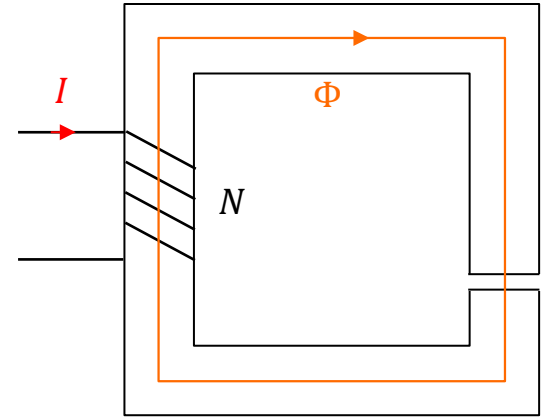
3.5 Magnetic circuit

3.5 Magnetic circuit

From magnetization to magnetic circuits



$$\vec{B} = \mu_0(\vec{H} + \vec{M})$$
$$B = \frac{\Phi}{A}$$



- **Microscopic level:** magnetic materials respond through the alignment of magnetic dipoles and domains.
- **Macroscopic level:** we describe the same behaviour by the material relation

$$\vec{B} = \mu_0(\vec{H} + \vec{M})$$

- **Engineering models**, the material response is captured by an effective permeability μ or μ_r , so that

$$\vec{B} = \mu \vec{H} = \mu_0 \mu_r \vec{H}$$

- This allows us to **replace the microscopic picture by a magnetic-circuit model** with flux Φ , field intensity H , and magnetomotive force (Ampère's circuital law).

3.5 Magnetic circuit

About magnetic circuits

- Magnetic circuits are path of **confined magnetic flux** Φ .
- ┐ Iron core: high permeability guiding the magnetic flux.
- ┐ Air gap: low permeability (high reluctance for the magnetic flux) to avoid saturation.
- **Magnetic flux** Φ is **constant** and independent of the material.
- **Losses** due to stray fields are **neglected**.
- The **magnetic flux density** depends on the **area** of the path.

$$B = \frac{\Phi}{A}$$

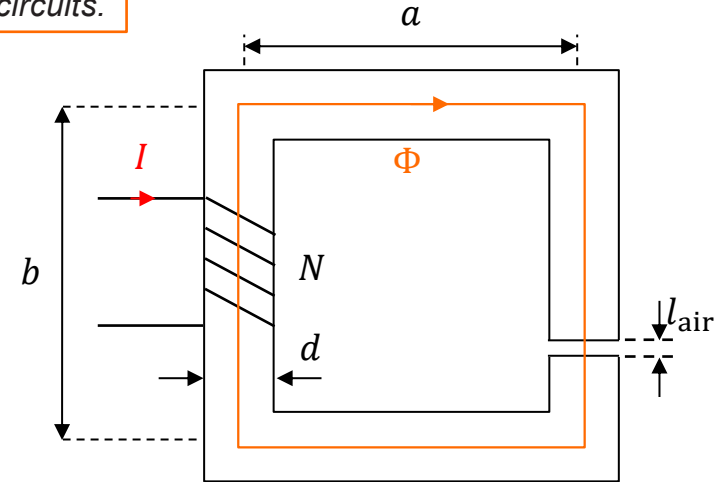
- The **magnetic field** H depends on the **permeability** μ_r .

$$H_{\text{core}} = \frac{B}{\mu_{r,\text{core}} \mu_0} \quad \text{and} \quad H_{\text{air}} = \frac{B}{\mu_0}$$

- The **current** I required to **drive the flux through the circuit**, follows **Ampère's circuital law** for all path-elements i .

$$\sum_i H_i l_i = N I$$

Analogue to the current I in electric circuits.



In an air gap, the magnetic field intensity increases by a factor of hundreds or even thousands.

Magnetic circuit of length $l = 2(a + b)$, core diameter d , air gap s and the relative permeability μ_r .

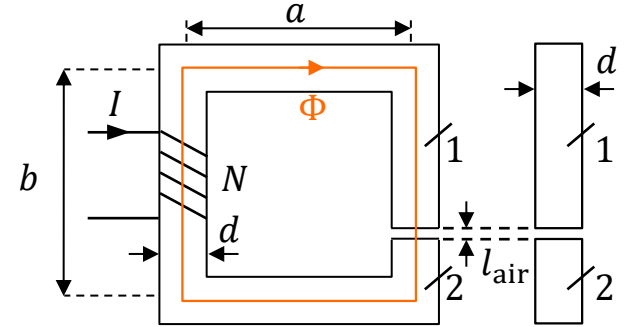
3.5 Magnetic circuit

Standard operation procedure

1. Find the magnetic flux Φ through the circuit.
2. Calculate the magnetic flux density $B = \Phi/A$
3. Calculate the current (MMF) required to drive the flux by adding the magnetic fields H_i of all length elements l_i

$$\sum_i H_i l_i = N I$$

- *Example: Given is a circuit with square cross-section of area $A = d^2$, a core length l_{core} , an air gap l_{air} , a permeability μ_r . Find the current I through N windings to generate a flux Φ .*



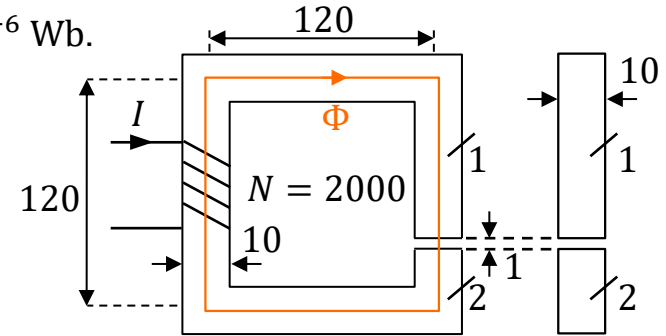
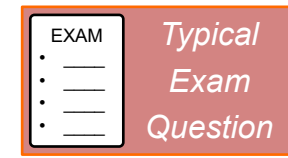
Left: Magnetic circuit of length $l_{\text{core}} = 2(a + b)$, core diameter d , air gap l_{air} and the relative permeability μ_r .

Right: Cross section.

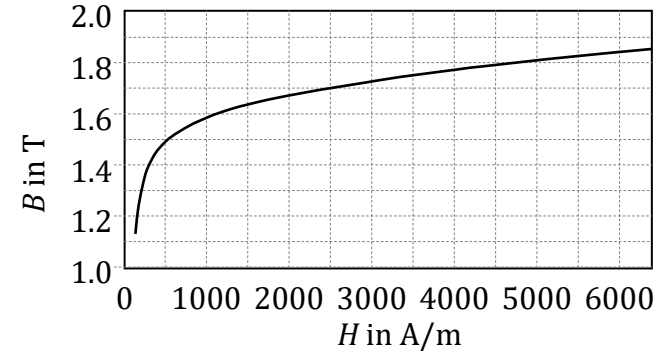
3.5 Magnetic circuit

Example: Magnetic circuit with air gap

Calculate the current required to produce a magnetic flux $\Phi = 170 \cdot 10^{-6}$ Wb.



Magnetic circuit. All dimensions in mm.



Magnetisation curve of sheet steel.

Summary

Magnetic circuits can be considered the magnetic equivalent to electric circuits, where

- The magnetic **flux** is the equivalent to the electric **current**.
- The magnetomotive **force** is the equivalent to the electric **voltage**.
- **Air gaps** with low permeability are the equivalent to **resistors**.

In magnetic circuits ...

- the magnetic **flux** is **constant**.
- the magnetic **flux density** only **depends** on the **flux** through and the **area** of the circuit.
- the magnetic **field intensity** is **inversely proportional** to the **relative permeability**.

3.5 Magnetic circuit

Learning objective

Students will be able to:

- Explain Maxwell's second equation / Faraday's law of induction.
- Calculate the voltage induced by a changing magnetic field through a loop of wire.
- Describe the self induction of an electrical conductor.
- Calculate the energy stored in a magnetic field.

- *Note: When talking about currents and voltages in this chapter, they might be constant or time varying, dependent of the rate of change of the magnetic flux. We therefore utilize small letters i and u .*

3.6 Time varying fields / Electromagnetic induction

3.6 Electromagnetic induction

Faraday's law of induction 1/2

"A time-varying magnetic field produces an eddy field that opposes the change in the magnetic flux."

– Maxwell's 2nd / Faraday's law of induction / Lenz law

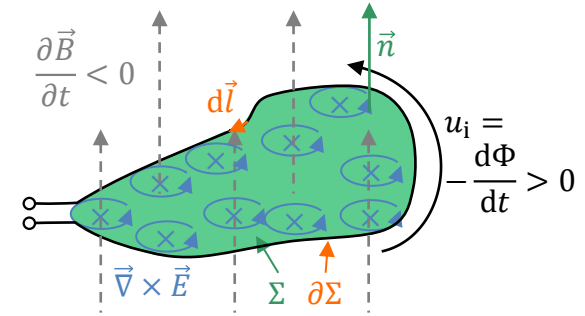
- In differential form

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

- In integral form for a loop $\partial\Sigma$ bounding the surface Σ ,

$$\oint_{\Sigma} \vec{\nabla} \times \vec{E} \cdot d\vec{A} = -\frac{\partial}{\partial t} \underbrace{\oint_{\Sigma} \vec{B} \cdot d\vec{A}}_{\text{Magnetic flux } \Phi} = \underbrace{\oint_{\partial\Sigma} \vec{E} \cdot d\vec{l}}_{\text{Induced voltage } u_i}$$

- In other words: when the electric field acts along a conducting loop, it induces a voltage u_i in the loop.



A wire placed in a time-changing magnetic field will induce a voltage between its two terminals.



The orientation of the path $\partial\Sigma$ and the surface normal $\vec{n}$ must be chosen consistently with the right-hand grip rule. Reversing the direction of the path also reverses the surface normal and changes the sign of u_i . Decreasing flux density $\partial\vec{B}/\partial t < 0$ in direction of $\vec{n}$ means decreasing flux, and vice versa. Increasing flux density $\partial\vec{B}/\partial t > 0$ in opposite to $\vec{n}$ means decreasing flux, and vice versa.

3.6 Electromagnetic induction

Faraday's law of induction 2/2

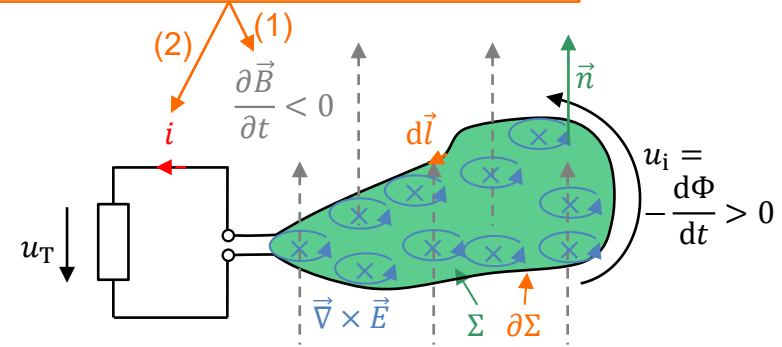
- For N turns of wire in an inductor the induced voltage u_i is:

$$u_i = -N \frac{d\Phi}{dt} = -N \frac{d}{dt} \vec{B} \cdot \vec{A}$$

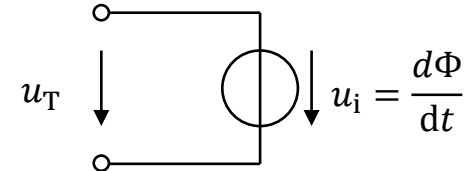
- The reason for **changing magnetic flux** $\frac{d\Phi}{dt}$ can be:

- the flux density $\vec{B}$ changes, for example in a transformer
- the loop area $\vec{A}$ changes because the wire loop is deformed;
- the angle between $\vec{B}$ and $\vec{A}$ changes, for example when a loop rotates in a magnetic field;
- or several of these effects occur at the same time.
- When the circuit is closed, the induced voltage drives a current according to the passive sign convention.
- In a lumped-element circuit model, induction can be represented by an equivalent voltage source whose polarity opposes the flux change.

When the magnetic field in direction $\vec{n}$ weakens (1), the induced voltage drives a current (2) that counteracts this change in magnetic flux (not illustrated), as stated by Lenz's law.



A wire placed in a time-changing magnetic field will induce a voltage between its two terminals.



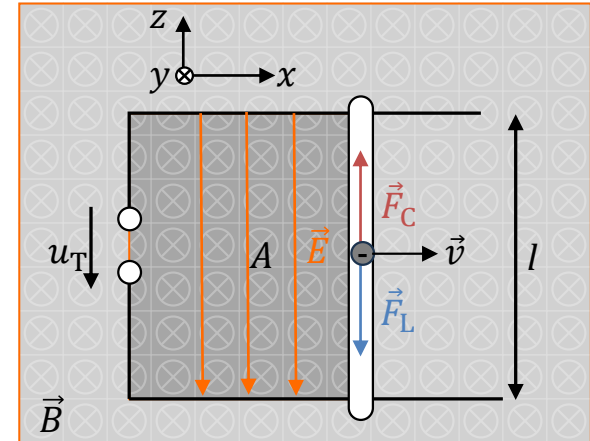
The induced voltage can be modelled as a voltage source u_i using active sign convention!

3.6 Electromagnetic induction

Example 1/2: Induction calculated via movement of a conductor

EXAM	Typical
• —	Exam
• —	Question
• —	

Calculate the terminal voltage u_T of a straight wire of length $l = 10$ cm moving at a velocity $\|\vec{v}\| = 10$ m/s perpendicular to a homogenous and static magnetic flux density $\|\vec{B}\| = 0.1$ T via the Lorentz's force.



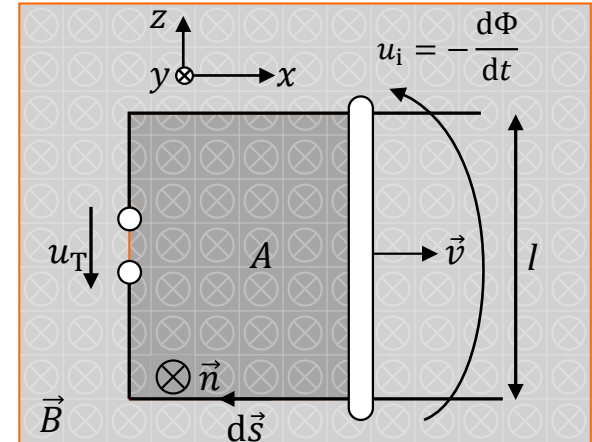
A wire moving with velocity $\vec{v}$ perpendicular to a magnetic flux density $\vec{B}$.

3.6 Electromagnetic induction

Example 2/2: Induction calculated via change of flux

EXAM	Typical
•	Exam
•	Question
•	
•	

Calculate the terminal voltage u_T of a straight wire of length $l = 10$ cm moving at a velocity $\|\vec{v}\| = 10$ m/s perpendicular to a homogenous and static magnetic flux density $\|\vec{B}\| = 0.1$ T via Faraday's law.



A wire moving with velocity $\vec{v}$ perpendicular to a magnetic flux density $\vec{B}$

3.6 Electromagnetic induction

Self induction

- When a loop of wire is connected to a voltage source u_s a current will start flowing.
- The rising current $\frac{di}{dt}$ generates increasing magnetic flux $\frac{d\Phi}{dt}$ according to Ampère's circuital law
- The ratio of linked flux over current is called inductance L

$$L = \frac{N \Phi}{i}$$

- The changing magnetic flux induces a voltage u_L across the inductor opposing the increase of current.

$$u_L = N \frac{d\Phi}{dt} = L \frac{di}{dt}$$

- *Note on sign convention: In this setup, the loop acts as a **load**, so we use the passive sign convention (Verbraucherzählpfeilsystem): the reference arrow of u_L points into the terminal where the current enters the inductor.*

For real applications, a resistor must be added to avoid short circuit under DC conditions.

The inductor acts as voltage source.

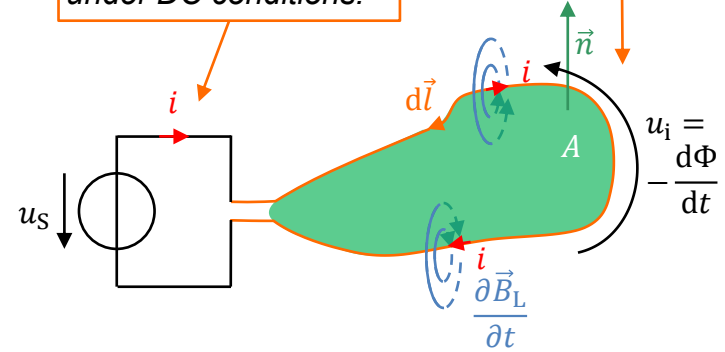
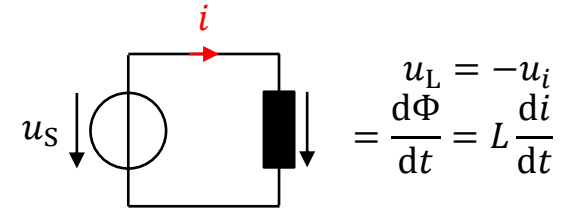


Illustration of the induced voltage u_L according to **active sign convention** (i opposite to u_L).



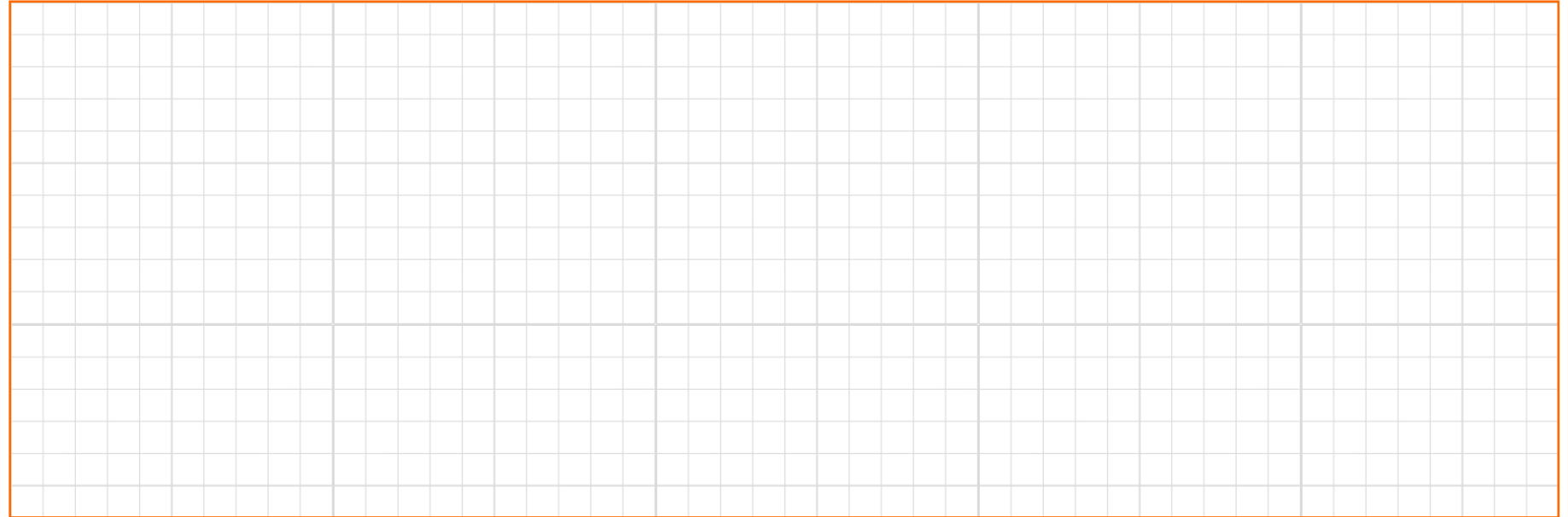
Representation of the field behaviour as **passive element** (i in direction of u_L).

3.6 Electromagnetic induction

Homework: Faraday's law of induction

EXAM	Typical
•	Exam
•	Question
•	
•	

A closed coil of $N = 50$ turns and area $A = 150 \text{ cm}^2$ is rotated in a magnetic field of flux density $\|\vec{B}\| = 2 \text{ Wb/m}^2$. It rotates from a position where its plane makes an angle $\theta = 45^\circ$ with the field to a position perpendicular to the field in a time $t = 0.3 \text{ s}$. Find the absolute value of the induced voltage $|u_i|$ in the coil due to its rotation.

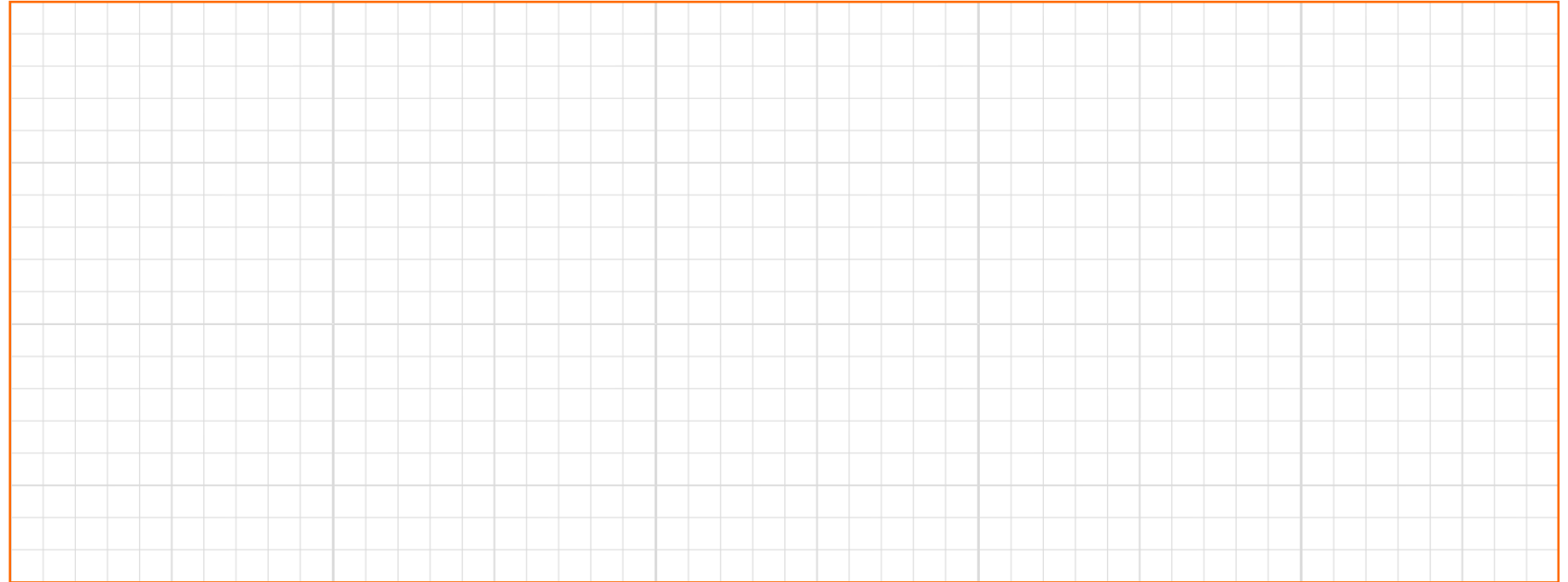


3.6 Electromagnetic induction

Homework: Faraday's law of induction

EXAM	Typical
• —	Exam
• —	Question
• —	

A uniform magnetic field is normal to the plane of a circular loop $D = 15$ cm in diameter with the resistance $R = 1.1$ m Ω . Calculate the rate of change of the magnetic flux density if the induced current is $I = 15$ A?



3.6 Electromagnetic induction

Inductance



Even a single winding of infinite radius a.k.a an infinitely long straight wire makes an inductor.

- Any arrangement of a current-carrying winding that produces a magnetic field is called an inductor (coil).
- The proportionality between magnetic flux linkage $N \Phi$ and total current I is the inductance L

$$L = \frac{N \Phi}{I}, \quad [L] = \frac{\text{Vs}}{\text{A}} = \text{H}$$

- For special geometries, where $d\vec{A} \parallel \vec{B}$, $d\vec{l} \parallel \vec{H}$ and $\|\vec{H}\|$ constant along the loop, the inductance can be written as

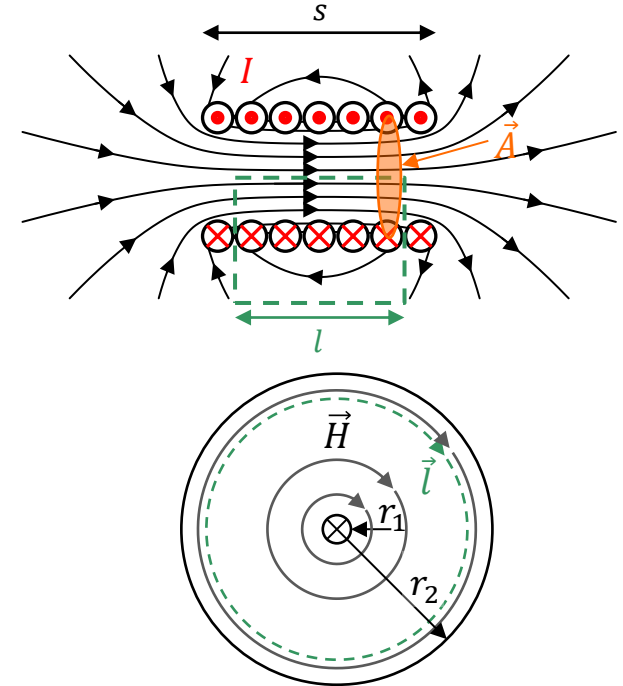
$$L = \frac{\mu N}{I} \oint \|\vec{H}\| dA = \mu N^2 \oint \frac{1}{l} dA$$

- Examples (constant $\mu = \mu_0 \mu_r$)

- Solenoid and toroid with (mean) length s , area A)

$$\|\vec{B}\| = \mu \frac{N}{s} I, \quad L = \mu \frac{N^2 A}{s}$$

- Coaxial cable (length s , inner r_1 , outer r_2): $L = \frac{\mu s}{2\pi} \ln \frac{r_1}{r_2}$

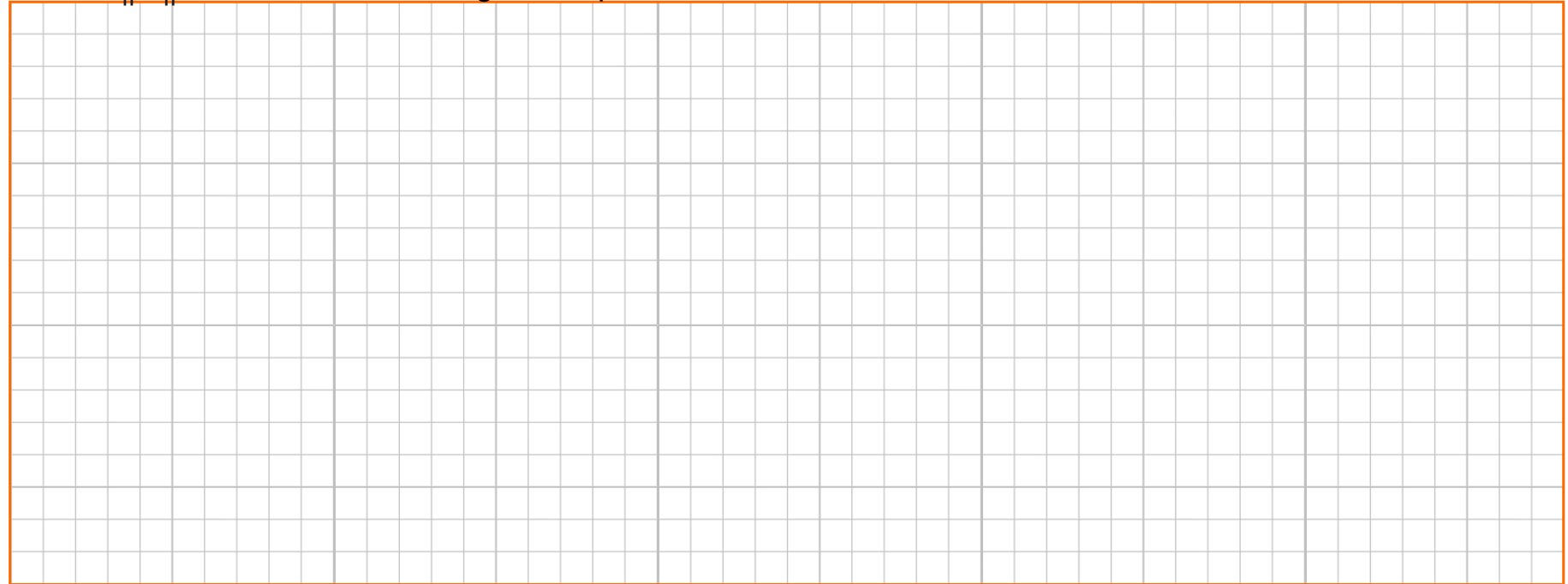


Schematic of a solenoid (top) and coaxial cable (bottom) including field intensity.

3.6 Electromagnetic induction

Homework: inductance for special geometries

Demonstrate that $L = \frac{\mu N}{l} \oint \|\vec{H}\| \, dA$ applies for geometries where $d\vec{A} \parallel \vec{B}$, $d\vec{l} \parallel \vec{H}$ and that $L = \mu N^2 \oint \frac{1}{l} \, dA$ holds if $\|\vec{H}\|$ is also constant along the loop l .

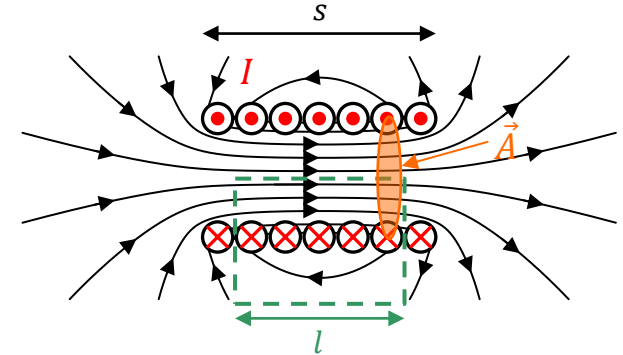


3.6 Electromagnetic induction

Homework: Inductance of a solenoid

EXAM	Typical
•	Exam
•	Question
•	
•	

Calculate the inductance per unit length of a long solenoid of length s of area A and N windings.
by applying Faraday's law of induction

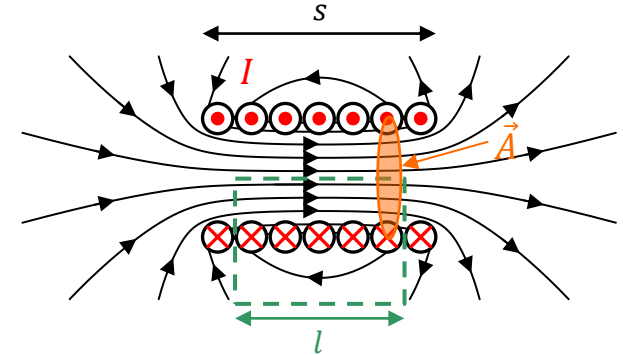


3.6 Electromagnetic induction

Homework: Inductance of a solenoid

EXAM	Typical
•	Exam
•	Question
•	
•	

Calculate the inductance per unit length of a long solenoid of length s of area A and N windings by directly applying the definition of the inductance.



3.6 Electromagnetic induction

Energy stored in the magnetic field

- Inductors store electric energy in the magnetic field; the associated state variable is the magnetic flux Φ (or flux linkage $N\Phi$).
- By analogy with the capacitor and using the definition of self-inductance $\Phi = LI$, the incremental work is
$$dW = i d\Phi = L i di$$
- Integrating from 0 to I and using $\Phi = LI$ gives the energy of an inductor:

$$W = \frac{1}{2} L I^2 = \frac{1}{2} \frac{\Phi^2}{L} = \frac{1}{2} \Phi I \quad [W] = \text{J}$$

- For a solenoid with uniform field, the magnetic energy density (work per unit volume) is

$$w_V = \frac{1}{2} \mu \|\vec{H}\|^2$$

Summary

- A **time varying magnetic flux** generates an **electric eddy field**.
- The eddy field induces a voltage along the wire according to $u_i = -\frac{d\Phi}{dt}$.
- N loops of wire in a changing magnetic flux it **induces a voltage** $u_i = -N \frac{d\Phi}{dt}$.
- Strictly follow the right-hand grip rule do correctly orient the surface and the path around the loop.
- **Inductance** $L = \frac{\Phi}{i}$ is the **ability to store magnetic flux** (analogue to the capacitance).
- The **energy** stored in a magnetic field is **proportional to the inductance** and the **current squared**.

3.6 Electromagnetic induction

Learning objective

Students will be able to:

- Explain the working principle of an ideal transformer.
- Apply your knowledge on electromagnetic induction and Ampère's circuital law.
- Apply the dot convention to define the phase of the secondary circuit.
- Calculate the parameters (voltage, current, turns ratio, reflected impedance) of ideal transformer.

3.7 Ideal transformer

3.7 Ideal transformer

Working principle of an ideal transformer 1/2

- A transformer **transmits electrical power** through free space!
- Alternating voltage in **primary winding** induces changing **magnetic flux** proportional to

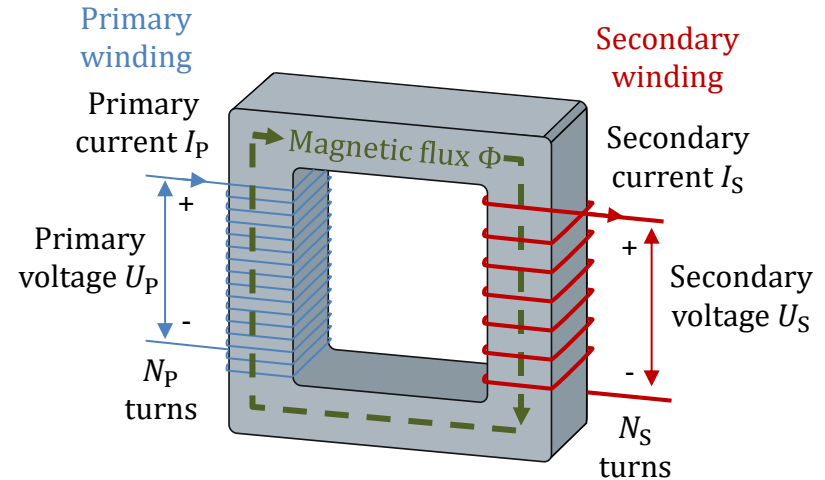
$$U_P = N_P \frac{d\Phi}{dt}$$

- Same magnetic flux enters **secondary winding** and **generates a voltage** U_S between terminals proportional to

$$U_S = N_S \frac{d\Phi}{dt}$$

- **Note:**

- 7 The current direction of second coil might be counter-intuitive. **Remember Lenz-rule:** The current is flowing so that it opposes the original magnetic field.



Schematic of a transformer.

3.7 Ideal transformer

Working principle of an ideal transformer 2/2

- A transformer **does not change** apparent power

$$|\underline{S}_P| = |\underline{S}_S|$$
$$U_P I_P = U_S I_S$$

- The induced voltage is **proportional** to the **number of wires N**
- The **relation** between the **primary** and the **secondary** winding is

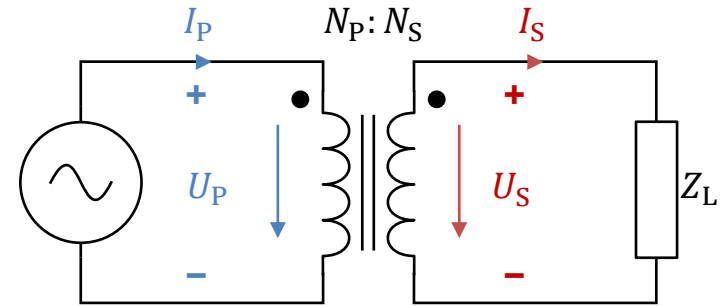
⌈ Voltage:

$$\frac{U_P}{U_S} = \frac{N_P}{N_S}$$

⌈ Current:

$$\frac{I_P}{I_S} = \frac{N_S}{N_P}$$

- **Number of turns in primary smaller** than number of turns in secondary → **Step up**
- **Number of turns in primary larger** than number of turns in secondary → **Step down**



Circuit diagram of an ideal transformer.

3.7 Ideal transformer

Dot convention

- The phase relation depends on the direction of the wrapping of the windings.
- The **dot convention** is used to indicate the **phase relation** between primary and secondary voltages and currents.
- Case 1: dots on the **same side**:
 - ⌋ If the current in coil 1 enters the dot, the mutually-induced voltage across coil 2 is positive.
 - ⌋ The voltage is **in phase**.
- Case 2: dots on **opposite side**:
 - ⌋ If the current in coil 1 enters the dot, the mutually induced voltage across coil 1 is negative.
 - ⌋ The voltage is **out of phase**.
- *Note: In general transformers are in phase.*

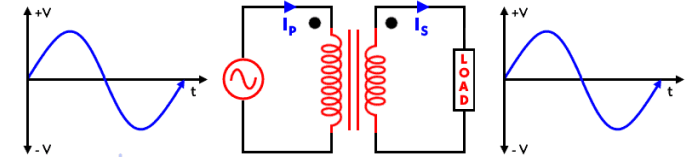


Fig a: Same Dots = Voltage are in Phase

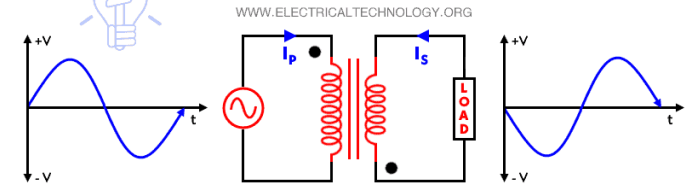


Fig b: Opposite Dots = Voltage are Out of Phase

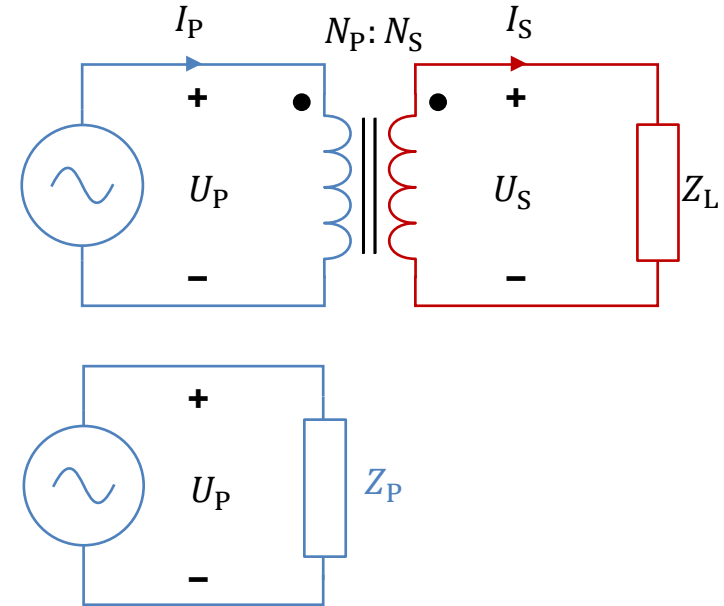
Schematic of the dot convention to define the phase and current of the secondary circuit.

3.7 Ideal transformer

Reflected impedance

- The **current** I_P generates a **magnetic flux** Φ_P in the **primary circuit** that **induces a voltage** U_S in the **secondary circuit**.
- The **current** I_S generates a **magnetic flux** Φ_S in the **secondary circuit** that **induces a voltage** U_P in the **primary circuit**.
- Even though, the **primary circuit is not physically connected** to the **load** Z_L in the secondary it **adjusts to the current** I_P in the primary accordingly → **reflected impedance** Z_P .
- With the voltage and current dependence on the turns ration the **reflected impedance** Z_P is:

$$Z_P = \frac{U_P}{I_P} = \frac{\frac{N_P}{N_S} U_S}{\frac{N_S}{N_P} I_S} = \left(\frac{N_P}{N_S} \right)^2 Z_L$$



Top: Circuit diagram of an ideal transformer.

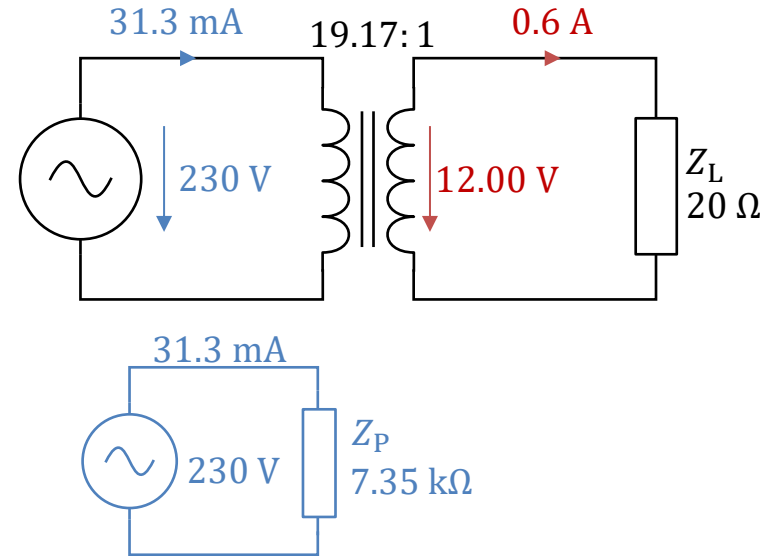
Bottom: Equivalent circuit for the reflected impedance in the primary circuit.

3.7 Ideal transformer

Example: Ideal transformer

EXAM	Typical
•	Exam
•	Question
•	
•	

Consider an ideal transformer with a turns ration of 19.17 to 1. What is the voltage in the secondary circuit if the input voltage is 230 V? What is the current of the primary I_P and secondary circuit I_S ? Calculate the reflected impedance Z_P as seen by the primary circuit.

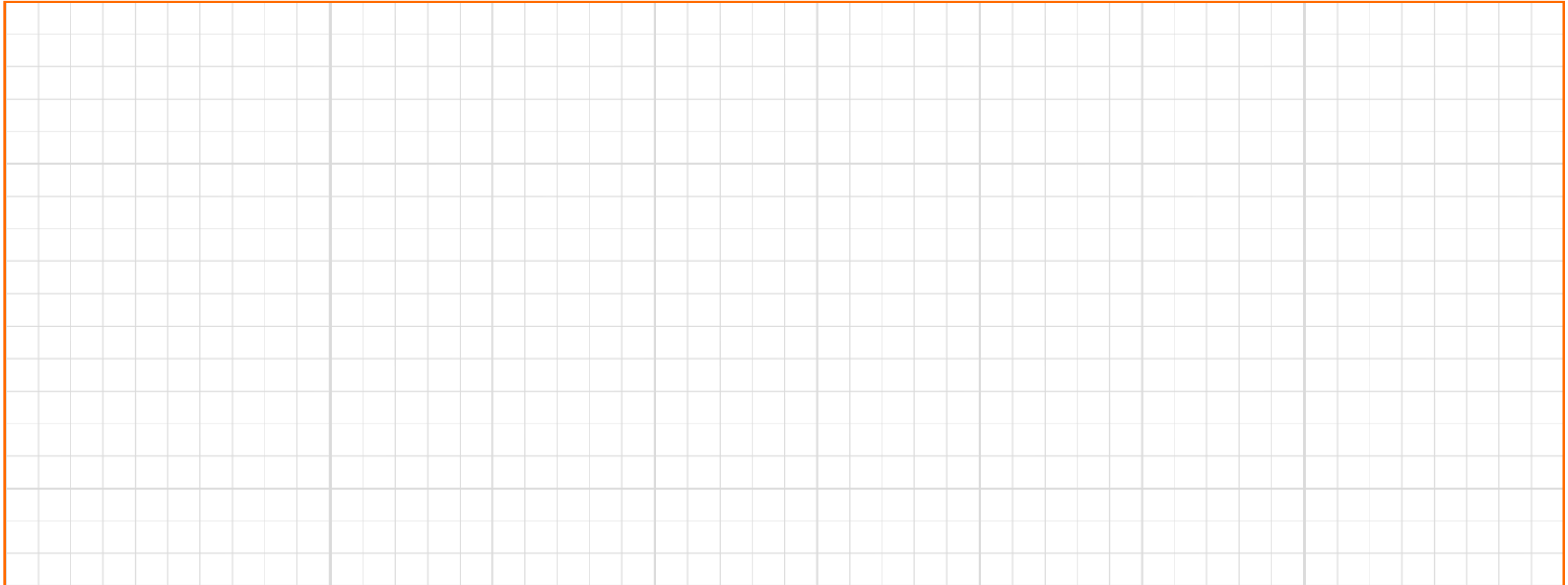


3.7 Ideal transformer

Exercise: Power transmission 1/2

EXAM	Typical
• —	Exam
• —	Question
• —	
• —	

Calculate the power lost in a transmission line without transformer.

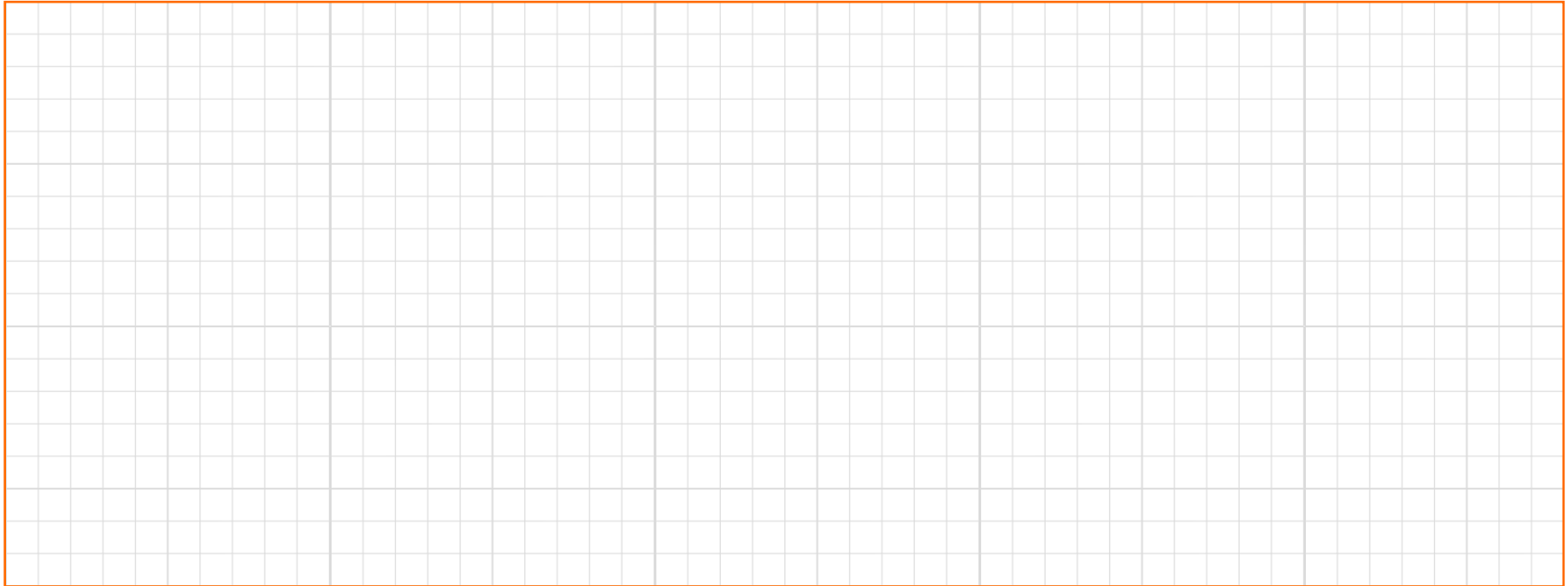


3.7 Ideal transformer

Exercise: Power transmission 2/2

EXAM	Typical Exam Question
• —	
• —	
• —	

Now calculate the power lost in a transmission line when transforming the voltage, first.



Summary

- Transformer **transfer electric power** from the **primary** to the **secondary** winding.
- An **ideal transformer** ...
 - ┐ does **not change** apparent **power**.
 - ┐ does **change** the **voltage ratio proportional** to the **turns ratio**.
 - ┐ does **change** the **current ratio inversely proportional** to the **turns ratio**.
- The **dot convention** is used to indicate the **phase relation** between primary and secondary circuit.

3.7 Ideal transformer

Vector calculus

Lower case greek letters & commonly used capitals

α	alpha
β	beta
$\gamma \Gamma$	gamma
$\delta \Delta$	delta
$\epsilon (\varepsilon)$	epsilon ('varepsilon')
ζ	zeta
η	eta
$\theta \Theta$	theta

ι	iota
κ	kappa
$\lambda \Lambda$	lambda
μ	mu
ν	nu
$\xi \Xi$	xi
$\omicron$	omicron
$\pi \Pi$	pi

ρ	rho
$\sigma \Sigma$	sigma
τ	tau
$\upsilon \Upsilon$	upsilon
$\phi (\varphi) \Phi$	phi ('varphi')
χ	chi
$\psi \Psi$	psi
$\omega \Omega$	omega

Version 4.0.0

- Initial version SS26

Version 4.1.1

- Solution path of exercise Biot-Savart extended, and nomenclature aligned to lecture slide.

Version 4.2.0

- Changed the orientation of the flux density with respect to the surface normal on slide 47 for better orientation. Symbols in ch. 4.7 changed in accordance with IRO1/BEE, slide relating Gauss' law to Ampère's circuital law on p. 37 added, slides on inductance aligned to the slides on capacitance

Version 4.3.0

- Hint for exam relevant questions added